#1 · Evaluation

How is MCV calculated from the hematocrit and the red cell count?

Answer

MCV (fL) = hematocrit (L/L) x 1000 divided by RBC count (x10^12/L). Example: Hct 0.40 and RBC 4.0 gives 100 fL. MCV classifies anemia as microcytic, normocytic or macrocytic.

#2 · Evaluation

What does a high RDW tell you about the red cell population?

Answer

A high RDW means marked anisocytosis, a mix of cell sizes. It is typical of early iron deficiency and megaloblastic anemia, and a normal RDW fits thalassemia trait or chronic disease.

#3 · Evaluation

How is MCHC calculated and which condition characteristically raises it?

Answer

MCHC (g/L) = hemoglobin (g/L) divided by hematocrit (L/L). It is raised in hereditary spherocytosis because spherocytes lose membrane and become dense. A falsely high MCHC also suggests cold agglutinins or lipemia.

#4 · Evaluation

What is the formula for MCH and what does a low value mean?

Answer

MCH (pg) = hemoglobin (g/L) divided by RBC count (x10^12/L). A low MCH means hypochromia, seen with iron deficiency and thalassemia, where cells carry too little hemoglobin.

#5 · Evaluation

In a stepwise anemia work-up, which two results sort most anemias first?

Answer

Start with MCV (microcytic, normocytic, macrocytic), then the reticulocyte count. A high retic count means bone marrow response, such as blood loss or hemolysis. A low count means production failure.

#6 · Evaluation

A patient has normocytic anemia with a low reticulocyte count. What does that suggest?

Answer

It suggests a hypoproliferative anemia: the marrow is not responding. Think chronic disease, renal disease, early iron deficiency or marrow failure. Hemolysis and bleeding would raise the retic count.

#7 · Evaluation

What is the corrected reticulocyte count and why is it used?

Answer

Corrected retic % = retic % x (patient Hct divided by normal Hct). It adjusts for the lower red cell mass in anemia, so a raw retic % does not overstate marrow response.

#8 · Evaluation

What is the typical CBC pattern in iron deficiency anemia?

Answer

Low hemoglobin, low MCV, low MCH, high RDW, and often a raised platelet count. The red cell count is relatively reduced for the degree of anemia, and the film is hypochromic.

#9 · Evaluation

Which blood film features are typical of iron deficiency anemia?

Answer

Hypochromic microcytic cells with marked central pallor, pencil cells (elongated elliptocytes) and anisopoikilocytosis. Target cells may be present but are fewer than in thalassemia.

#10 · Evaluation

What is the classic iron study pattern in iron deficiency?

Answer

Low serum iron and ferritin, high total iron binding capacity (transferrin), and low transferrin saturation. Ferritin is the earliest and most specific marker of depleted stores.

#11 · Evaluation

How do iron studies differ in anemia of chronic disease compared with iron deficiency?

Answer

In chronic disease, serum iron is low and TIBC is low or normal, with normal or high ferritin because ferritin is an acute phase reactant. In iron deficiency, TIBC is high and ferritin is low.

#12 · Evaluation

What is the Mentzer index and how does it separate thalassemia trait from iron deficiency?

Answer

Mentzer index = MCV (fL) divided by RBC count (x10^12/L). A value under 13 suggests thalassemia trait (high RBC count, low MCV). A value over 13 suggests iron deficiency.

#13 · Evaluation

Why is the RBC count usually high relative to hemoglobin in thalassemia trait?

Answer

Thalassemia trait produces many small cells, so the RBC count stays normal or high despite a mild anemia and low MCV. Iron deficiency lowers the RBC count along with the MCV.

#14 · Evaluation

Which blood film findings are typical of megaloblastic anemia?

Answer

Macro-ovalocytes, anisopoikilocytosis, hypersegmented neutrophils (five or more lobes) and sometimes Howell-Jolly bodies. The MCV is often over 110 fL, and the retic count is low.

#15 · Evaluation

What is the usual cause of the high MCV in vitamin B12 and folate deficiency?

Answer

Impaired DNA synthesis delays nuclear maturation while cytoplasm keeps developing, producing large cells (nuclear-cytoplasmic asynchrony). The marrow is hypercellular with ineffective erythropoiesis, causing a raised LD and indirect bilirubin.

#16 · Evaluation

Which test helps separate B12 deficiency from folate deficiency, and what is the key difference?

Answer

Methylmalonic acid is raised in B12 deficiency but normal in folate deficiency. Homocysteine is raised in both. Neurologic symptoms point to B12 deficiency.

#17 · Evaluation

Which laboratory results together suggest hemolysis?

Answer

High reticulocyte count, raised LD and indirect bilirubin, low or absent haptoglobin, and polychromasia on the film. Intravascular hemolysis also gives hemoglobinuria and hemosiderinuria.

#18 · Evaluation

Why is haptoglobin low in hemolytic anemia?

Answer

Haptoglobin binds free hemoglobin and the complex is cleared by the liver. Hemolysis uses up haptoglobin faster than it is made. It is an acute phase reactant, so inflammation can mask the drop.

#19 · Evaluation

What do spherocytes on the film suggest and what pair of tests follows?

Answer

Spherocytes suggest immune hemolysis or hereditary spherocytosis. The direct antiglobulin test separates them: positive means autoimmune hemolytic anemia, negative favours hereditary spherocytosis.

#20 · Evaluation

Which lab pattern is characteristic of hereditary spherocytosis?

Answer

Spherocytes without central pallor, raised MCHC, raised retic count, a negative DAT and increased osmotic fragility. The eosin-5-maleimide (EMA) binding test is a common modern confirmation.

#21 · Evaluation

In what condition are elliptocytes the dominant cell, and what is the defect?

Answer

Hereditary elliptocytosis, caused by defects in red cell membrane skeletal proteins such as spectrin or protein 4.1. Many patients are asymptomatic, and the film shows numerous elliptocytes.

#22 · Evaluation

Which inclusions and cells point to oxidant hemolysis in G6PD deficiency?

Answer

Heinz bodies (visible only with supravital stains like crystal violet), bite cells and blister cells. Heinz bodies are denatured hemoglobin, and the spleen removes them to make bite cells.

#23 · Evaluation

Why can a G6PD screening test be falsely normal during or just after a hemolytic crisis?

Answer

G6PD-deficient older cells are destroyed first and young reticulocytes have higher enzyme activity. Testing should wait about two to three months after the episode to avoid a falsely normal result.

#24 · Evaluation

What does the sickle solubility test detect and what is its limitation?

Answer

It detects hemoglobin S, which is insoluble in reduced state and makes the solution turbid. It cannot distinguish sickle cell trait from disease, so hemoglobin electrophoresis or HPLC is needed to confirm.

#25 · Evaluation

What hemoglobin pattern is typical of beta thalassemia trait on electrophoresis or HPLC?

Answer

Raised HbA2 (usually above 3.5 percent), with HbF normal or mildly raised. The CBC shows low MCV and MCH with target cells. Co-existing iron deficiency can lower HbA2 and mask the pattern.

#26 · Evaluation

What HbA2 level typically supports a diagnosis of beta thalassemia trait on HPLC or electrophoresis?

Answer

HbA2 is raised, usually above 3.5 percent (often 4 to 6 percent), with or without a mild rise in HbF. Pair it with microcytosis and a normal or high red cell count.

#27 · Evaluation

On hemoglobin electrophoresis, how do sickle cell trait and sickle cell disease differ in their HbA and HbS bands?

Answer

Trait shows HbA plus HbS, with HbS lower than HbA (usually about 35 to 45 percent). Disease (HbSS) shows mostly HbS, no HbA, a raised HbF and a normal HbA2.

#28 · Evaluation

Which variants co-migrate with HbA2 on alkaline electrophoresis, and how can they be separated?

Answer

HbC, HbE and HbO-Arab all migrate with HbA2 at alkaline pH. Citrate agar (acid) electrophoresis or HPLC separates them and avoids misreading an HbA2 value.

#29 · Evaluation

A child shows mostly HbF with no HbA on HPLC and no transfusion history. What does this suggest?

Answer

Beta thalassemia major, where beta chain production is absent, so HbF is often 90 percent or more. Confirm with age, transfusion history, red cell indices and family studies.

#30 · Evaluation

List common causes of neutropenia.

Answer

Viral infection, drugs (chemotherapy, clozapine, antithyroid drugs), marrow failure, autoimmune disease and overwhelming sepsis. Some people also have benign ethnic neutropenia, a lower normal neutrophil count without increased infection risk.

#31 · Evaluation

In warm autoimmune hemolytic anemia, what is the typical direct antiglobulin test (DAT) result?

Answer

DAT is positive with anti-IgG, with or without anti-C3d. The film shows spherocytes and polychromasia, and the reticulocyte count is raised.

#32 · Evaluation

What DAT pattern and film finding suggest cold agglutinin disease?

Answer

DAT is positive with anti-C3d only (IgG negative). The film shows red cell agglutination. Warm the sample to 37 C before counting, since agglutinates falsely raise the MCV and lower the RBC count.

#33 · Evaluation

Which two conditions are the main causes of spherocytes, and what test separates them?

Answer

Hereditary spherocytosis (DAT negative, raised MCHC, increased osmotic fragility) and immune hemolytic anemia (DAT positive). The DAT is the first test that separates them.

#34 · Evaluation

How do reactive lymphocytes differ from malignant lymphocytes on a blood film?

Answer

Reactive cells are pleomorphic, with varied size, abundant cytoplasm and mature chromatin, giving a mixed population. Malignant lymphocytosis is monotonous, with one clone of similar looking cells.

#35 · Evaluation

How do coagulation results differ between TTP and DIC when schistocytes are present?

Answer

TTP has a normal PT and aPTT with a normal fibrinogen and severe thrombocytopenia. DIC has a prolonged PT and aPTT, low fibrinogen and raised D-dimer.

#36 · Evaluation

What blood film finding defines microangiopathic hemolytic anemia, and what is its mechanism?

Answer

Schistocytes (helmet cells, triangles, fragments) with polychromasia and thrombocytopenia, DAT negative. Red cells are sheared as they pass fibrin strands or platelet thrombi in small vessels.

#37 · Evaluation

Which enzyme deficiency causes TTP, and what activity level is typical?

Answer

ADAMTS13 activity below 10 percent, from an autoimmune inhibitor or a congenital defect. Without it, ultra large von Willebrand multimers persist and form platelet thrombi.

#38 · Evaluation

What defines aplastic anemia in the blood count and bone marrow?

Answer

Pancytopenia with a low reticulocyte count and a hypocellular marrow with fatty replacement, without abnormal infiltrate or fibrosis. Red cells are normocytic or mildly macrocytic.

#39 · Evaluation

Contrast marrow findings in pancytopenia from different causes.

Answer

Cellular marrow: megaloblastic anemia, myelodysplasia, hypersplenism. Hypocellular marrow: aplastic anemia. Infiltrated marrow: leukemia, lymphoma, metastases or myelofibrosis. The marrow biopsy separates these.

#40 · Evaluation

How is the reticulocyte count reported, and what is a typical adult reference range?

Answer

Reported as a percent of red cells, about 0.5 to 2.5 percent in adults (ranges vary by laboratory). Count at least 1000 red cells on a supravital stained film such as new methylene blue.

#41 · Evaluation

How do you calculate the corrected reticulocyte count?

Answer

Retic percent multiplied by (patient hematocrit divided by normal hematocrit). Example: 6 percent x (0.20 / 0.45) = 2.7 percent. It corrects for the reduced red cell number in anemia.

#42 · Evaluation

How is the reticulocyte production index calculated and interpreted?

Answer

RPI = corrected reticulocyte count divided by the maturation time in days (about 1.0 at hematocrit 0.45, 2.0 at 0.25). An RPI below 2 in anemia suggests an inadequate marrow response.

#43 · Evaluation

What distinguishes relative polycythemia from true polycythemia?

Answer

Relative: plasma volume falls (dehydration, diuretics), so hemoglobin and hematocrit rise with a normal red cell mass. True: red cell mass is actually increased, confirmed by measuring it.

#44 · Evaluation

How do EPO levels and JAK2 help separate polycythemia vera from secondary polycythemia?

Answer

Polycythemia vera: EPO low or low normal, JAK2 V617F present in about 95 percent. Secondary: EPO raised (hypoxia, EPO producing tumours) and JAK2 negative.

#45 · Evaluation

What are typical adult reference ranges for neutrophils and lymphocytes in a white cell differential?

Answer

Neutrophils about 40 to 70 percent (absolute 2.0 to 7.5 x10^9/L). Lymphocytes about 20 to 40 percent (absolute 1.0 to 4.0 x10^9/L). Ranges vary by laboratory.

#46 · Evaluation

How do you calculate an absolute cell count from a differential, and why does it matter?

Answer

Absolute count = WBC (x10^9/L) x percent of that cell / 100. Example: WBC 10 with 60 percent neutrophils gives 6.0 x10^9/L. Percentages alone can hide neutropenia or lymphocytosis.

#47 · Evaluation

What is a left shift on a blood film, and what usually causes it?

Answer

Increased band neutrophils, and sometimes metamyelocytes or myelocytes, in the blood. Usually from infection or inflammation, as the marrow releases younger neutrophils early.

#48 · Evaluation

Name the toxic changes seen in neutrophils during severe infection.

Answer

Toxic granulation, Dohle bodies (blue grey cytoplasmic RNA remnants), cytoplasmic vacuoles and sometimes giant bands. They reflect accelerated neutrophil production in the marrow.

#49 · Evaluation

What absolute neutrophil count defines severe neutropenia, and how does infection risk change?

Answer

An ANC below 0.5 x10^9/L is severe, with a high risk of serious bacterial infection. Neutropenia in adults is generally below 1.5 x10^9/L, and risk rises as the ANC falls.

#50 · Evaluation

What blood film pattern suggests infectious mononucleosis?

Answer

Absolute lymphocytosis with many large reactive lymphocytes, usually EBV driven, with a positive heterophile (monospot) test. The reactive cells have abundant basophilic cytoplasm that indents around neighbouring red cells.

#51 · Evaluation

Name three common causes of eosinophilia on a differential.

Answer

Allergy and asthma, parasitic (helminth) infection, and drug reactions. Others include atopic skin disease and some neoplasms such as Hodgkin lymphoma. Eosinophils are normally 1 to 4 percent of the white cells.

#52 · Evaluation

A persistent absolute basophilia should make you think of which group of disorders?

Answer

Myeloproliferative neoplasms, especially chronic myeloid leukemia. Basophils are normally under 1 percent, so a rising basophil count with immature granulocytes is a red flag to flag for review.

#53 · Evaluation

Which conditions are typically associated with a reactive monocytosis?

Answer

Chronic infections such as tuberculosis, recovery from neutropenia or chemotherapy, and chronic inflammatory disease. A persistent unexplained monocytosis may also point to chronic myelomonocytic leukemia.

#54 · Evaluation

How does the leukocyte alkaline phosphatase (LAP) score help separate a leukemoid reaction from CML?

Answer

The LAP score is high in a leukemoid reaction and low in chronic myeloid leukemia. Neutrophils in infection are activated, while CML neutrophils are malignant and have little LAP activity.

#55 · Evaluation

Which smear feature favours CML over a leukemoid reaction besides a low LAP score?

Answer

Basophilia (and often eosinophilia) alongside a full spectrum of immature granulocytes. A leukemoid reaction shows a left shift with toxic changes (toxic granulation, Dohle bodies, vacuolation) but no increase in basophils.

#56 · Evaluation

What is the usual blast threshold on a bone marrow or blood count for diagnosing acute leukemia?

Answer

20 percent blasts or more is the classic WHO threshold, although some recent classifications allow a diagnosis with lower blast counts when a defining genetic abnormality is present.

#57 · Evaluation

What are Auer rods and what do they tell you?

Answer

Auer rods are needle-like, red to purple inclusions of fused primary granules in the cytoplasm of myeloid blasts. Seeing one confirms myeloid lineage, meaning acute myeloid leukemia, not ALL.

#58 · Evaluation

Which cytochemical stain is positive in AML blasts and negative in ALL blasts?

Answer

Myeloperoxidase (MPO), along with Sudan black B. Lymphoblasts are MPO negative. This is a classic first step to separate AML from ALL before flow cytometry.

#59 · Evaluation

What are the typical morphology features of acute promyelocytic leukemia (APL) cells?

Answer

Abnormal promyelocytes with heavy purple granulation, often bundles of Auer rods (faggot cells), and a bilobed or folded nucleus. The hypogranular variant has a butterfly-shaped nucleus.

#60 · Evaluation

Why is suspected APL treated as a medical emergency in the lab?

Answer

APL carries a high risk of DIC and fatal bleeding. A suspected APL smear should be phoned to the physician urgently, since early treatment with all-trans retinoic acid greatly improves survival.

#61 · Evaluation

What chromosomal translocation defines acute promyelocytic leukemia?

Answer

t(15;17), which creates the PML::RARA fusion gene. Confirmation is by cytogenetics, FISH, or PCR, but a smear with suspicious promyelocytes should be reported urgently before confirmation.

#62 · Evaluation

What is the Philadelphia chromosome and which leukemia is it linked to?

Answer

The t(9;22) translocation producing the BCR::ABL1 fusion gene. It is the hallmark of chronic myeloid leukemia and is found in about 95 percent of cases.

#63 · Evaluation

What does a typical CML blood film in chronic phase look like?

Answer

Marked leukocytosis with a full range of granulocyte maturation (myelocytes, metamyelocytes, bands, segmented cells), few blasts, basophilia, and sometimes thrombocytosis. Myelocytes and neutrophils usually predominate.

#64 · Evaluation

What are smudge cells and in which leukemia are they typical?

Answer

Smudge (basket) cells are fragile lymphocytes crushed during smear preparation. They are characteristic of chronic lymphocytic leukemia, which shows many small mature lymphocytes.

#65 · Evaluation

What persistent absolute lymphocyte count supports a diagnosis of CLL?

Answer

A sustained monoclonal B lymphocytosis of 5 x10^9/L or more, confirmed by flow cytometry. The smear shows small mature lymphocytes with clumped chromatin and scant cytoplasm.

#66 · Evaluation

Which mutation is found in most patients with polycythemia vera?

Answer

JAK2 V617F in over 95 percent of cases. The disease presents with raised hemoglobin and hematocrit, often with elevated white cells and platelets, and low erythropoietin.

#67 · Evaluation

What smear and count finding is the hallmark of essential thrombocythemia?

Answer

A sustained, markedly raised platelet count (450 x10^9/L or more) with large or giant platelets on the film. Reactive causes such as iron deficiency or inflammation must be excluded.

#68 · Evaluation

What red cell shape is classic for primary myelofibrosis on the blood film?

Answer

Teardrop cells (dacrocytes), along with a leukoerythroblastic picture of nucleated red cells and immature granulocytes. Bone marrow fibrosis causes the marrow aspirate to be a dry tap.

#69 · Evaluation

What dysplastic changes in neutrophils suggest a myelodysplastic syndrome?

Answer

Hypogranular or agranular cytoplasm and hyposegmented nuclei (pseudo-Pelger-Huet cells). Dysplasia in at least 10 percent of cells of a lineage is the usual threshold.

#70 · Evaluation

What is a pseudo-Pelger-Huet cell?

Answer

A neutrophil with a bilobed, pince-nez shaped nucleus and condensed chromatin, acquired in MDS or after chemotherapy. The inherited Pelger-Huet anomaly looks the same but the cells function normally.

#71 · Evaluation

Which abnormal erythroid finding in the bone marrow marks the MDS subtype with ring sideroblasts?

Answer

Ring sideroblasts: erythroid precursors with iron-laden mitochondria ringing the nucleus on Prussian blue stain. They must make up at least 15 percent of erythroid precursors (5 percent with an SF3B1 mutation).

#72 · Evaluation

What causes rouleaux on a blood film and why is it prominent in multiple myeloma?

Answer

Excess serum proteins (especially monoclonal immunoglobulin) neutralize the red cell surface charge so cells stack like coins. The background of the film often stains blue from the high protein content.

#73 · Evaluation

Which cell is the malignant clone in multiple myeloma and how does it look?

Answer

The plasma cell: oval cell with eccentric nucleus, clock-face chromatin, deep blue cytoplasm, and a pale perinuclear hof (Golgi zone). Flame cells and Mott cells are variants seen with abnormal immunoglobulin.

#74 · Evaluation

How do you recognize atypical lymphocytes from lymphoma cells in peripheral blood?

Answer

Lymphoma cells are often monomorphic, with cleaved, folded or irregular nuclei, a size range from small to large, and basophilic cytoplasm. Reactive lymphocytes in infection are pleomorphic, with abundant cytoplasm that hugs neighbouring red cells.

#75 · Evaluation

What platelet count range is generally used as the adult reference interval, and what count defines thrombocytopenia?

Answer

The usual adult reference interval is about 150 to 400 x10^9/L. Thrombocytopenia is a count below 150 x10^9/L. Always confirm a low count on the smear before reporting.

#76 · Evaluation

Name three common causes of reactive thrombocytosis.

Answer

Iron deficiency, acute infection or inflammation, and recent splenectomy or surgery. These raise platelets secondarily, unlike essential thrombocythemia, which is a clonal marrow disorder.

#77 · Evaluation

How are microfilariae recognized on a stained blood film, and why is collection timing important for some species?

Answer

They are long, sheathed or unsheathed worms with nuclei in a column, seen in thick films. Wuchereria and Brugia show nocturnal periodicity, so blood is best collected at night.

#78 · Evaluation

What causes EDTA-dependent pseudothrombocytopenia, and what does the smear show?

Answer

An in-vitro antibody reacts with platelets in EDTA, causing clumping. The smear shows platelet clumps and the analyzer count is falsely low, so the patient is not truly thrombocytopenic.

#79 · Evaluation

What is platelet satellitism on a blood film?

Answer

Platelets ring around neutrophils, an EDTA-related artifact. The platelets are not counted correctly by the analyzer, giving a falsely low count. A citrate recount can resolve it.

#80 · Evaluation

How do you correct a spurious platelet count caused by EDTA clumping?

Answer

Recollect in sodium citrate (light blue tube), recount, and multiply the result by 1.1 to correct for the citrate dilution. Confirm the clumps are absent on the new film.

#81 · Evaluation

What are the typical smear and platelet features of immune thrombocytopenia (ITP)?

Answer

Isolated low platelet count with normal red cell and white cell findings, and often large platelets. Bleeding is mucocutaneous. It is a diagnosis made after other causes are excluded.

#82 · Evaluation

Which blood film finding is the hallmark of thrombotic thrombocytopenic purpura (TTP)?

Answer

Schistocytes (fragmented red cells) with marked thrombocytopenia. Associated findings are microangiopathic hemolytic anemia and raised LDH, caused by severe ADAMTS13 deficiency.

#83 · Evaluation

Heparin-induced thrombocytopenia usually appears how many days after starting heparin, and what is the key lab clue?

Answer

Typically 5 to 10 days after starting heparin, with a platelet fall of more than 50 percent from baseline. Confirm with PF4 antibody testing; paradoxical thrombosis is the main risk.

#84 · Evaluation

Which inherited platelet disorder lacks GPIIb/IIIa and shows absent aggregation to ADP and collagen?

Answer

Glanzmann thrombasthenia. Platelet count and size are normal, and ristocetin aggregation is normal because GPIb is intact. Fibrinogen binding fails, so there is no platelet to platelet bridging.

#85 · Evaluation

Bernard-Soulier syndrome lacks which glycoprotein, and what is the typical smear and ristocetin result?

Answer

GPIb/IX/V is deficient. Findings are giant platelets, thrombocytopenia and absent ristocetin aggregation that is not corrected by normal plasma, because platelets cannot bind von Willebrand factor.

#86 · Evaluation

What is the usual coagulation pattern in von Willebrand disease?

Answer

Prolonged APTT is possible because factor VIII is reduced, with a normal PT and platelet count. Ristocetin cofactor activity and VWF antigen are low in most types.

#87 · Evaluation

What is a Howell-Jolly body, and what does its presence suggest?

Answer

A small round DNA remnant (nuclear fragment) in a red cell, purple on Wright stain. It suggests absent or hypofunctioning spleen, because the spleen normally removes these inclusions.

#88 · Evaluation

What do Pappenheimer bodies contain, and which stain confirms them?

Answer

They contain iron granules and appear as small basophilic granules near the cell periphery on Wright stain. Prussian blue stain confirms iron. Seen in sideroblastic anemia and after splenectomy.

#89 · Evaluation

Basophilic stippling is caused by what, and name one classic association.

Answer

Aggregated ribosomal RNA in red cells that stains blue as fine or coarse dots. Classic associations are lead poisoning, thalassemia and other disorders of impaired hemoglobin synthesis.

#90 · Evaluation

Which red cell shape is associated with liver disease and hemoglobin C, appearing as a bullseye?

Answer

Target cells (codocytes), with a central dark spot ringed by a pale area. Also seen in thalassemia, iron deficiency and after splenectomy, due to excess membrane relative to hemoglobin.

#91 · Evaluation

How do acanthocytes differ from echinocytes?

Answer

Acanthocytes have a few irregular spicules of varying length. Echinocytes (burr cells) have 10 to 30 short, evenly spaced projections and are often an artifact or seen in uremia.

#92 · Evaluation

Teardrop cells (dacrocytes) are classically seen in which condition?

Answer

Myelofibrosis, where marrow fibrosis distorts cells as they leave. They also occur in thalassemia and myelophthisic anemia, usually with a leukoerythroblastic picture.

#93 · Evaluation

What is the central pallor shape of a stomatocyte?

Answer

A slit-like or mouth-shaped area of central pallor. Seen in hereditary stomatocytosis, alcoholism and liver disease, and sometimes as a drying artifact.

#94 · Evaluation

How is the corrected WBC count calculated when nucleated RBCs are present?

Answer

Corrected WBC = observed WBC x 100 divided by (100 + nRBC per 100 WBC). Example: WBC 10.0 with 25 nRBC per 100 WBC gives 10.0 x 100 / 125 = 8.0 x10^9/L.

#95 · Evaluation

What does polychromasia on a Wright-stained film indicate, and what supravital stain confirms reticulocytes?

Answer

Blue-grey larger red cells that are young reticulocytes with residual RNA, indicating marrow response. New methylene blue or brilliant cresyl blue shows the reticulum for reticulocyte counts.

#96 · Evaluation

On a thick malaria smear, what is its main advantage over a thin smear?

Answer

It concentrates parasites about 20 to 40 times, so it is more sensitive for detection. The thin smear is used for species identification and calculating percent parasitemia.

#97 · Evaluation

Which Plasmodium species shows banana-shaped gametocytes and multiple ring forms with applique forms?

Answer

Plasmodium falciparum. Only rings and gametocytes are normally seen in peripheral blood, and parasitemia can be very high. It is the species associated with severe, life-threatening malaria.

#98 · Evaluation

Which lab features help recognize Babesia, and what is it commonly mistaken for?

Answer

Tiny pleomorphic rings, sometimes in a Maltese cross tetrad form, with no pigment or gametocytes. It is often mistaken for P. falciparum rings, so travel and tick exposure history help.

#99 · Evaluation

In bone marrow, what is the normal myeloid to erythroid (M:E) ratio?

Answer

About 3:1 to 4:1 in the healthy adult marrow. Maturation is assessed from the least to most mature cell, from blast through to the segmented neutrophil or orthochromatic normoblast.

#100 · Evaluation

In the normal myeloid series, which is the last cell stage that can still divide?

Answer

The myelocyte is the last stage able to divide. Metamyelocytes, bands and segmented neutrophils are maturing cells only. Memory hook: the myelocyte is where secondary granules appear and the nucleus is round to oval.

#101 · Evaluation

How does a normoblast (orthochromatic erythroblast) differ from a polychromatic erythroblast on a Wright-stained film?

Answer

The orthochromatic erythroblast has a small, dense, pyknotic nucleus with no visible chromatin pattern and pink cytoplasm. The polychromatic erythroblast has coarser clumped chromatin and blue-grey cytoplasm from mixed hemoglobin and RNA.

#102 · Evaluation

What are the key features of a normal megakaryocyte in bone marrow, and what do platelets arise from?

Answer

A large cell (up to 160 um) with a multilobed nucleus and abundant granular cytoplasm. Platelets are shed as cytoplasmic fragments from the megakaryocyte. Thrombopoietin is the main growth factor driving its maturation.

#103 · Evaluation

A patient on warfarin has a prolonged PT. Which coagulation factor falls first and why does it drive the early PT rise?

Answer

Factor VII has the shortest half-life of the vitamin K dependent factors (about 6 hours), so it falls first. Because PT is sensitive to factor VII, the PT lengthens within the first days of therapy.

#104 · Evaluation

What does the INR correct for, and what formula is used to calculate it?

Answer

The INR corrects for differences in thromboplastin reagent sensitivity between labs. INR = (patient PT / mean normal PT) raised to the power of the ISI, where ISI is the reagent's international sensitivity index.

#105 · Evaluation

A patient not on anticoagulants has an isolated prolonged PT with a normal aPTT. Which factor deficiency is suspected?

Answer

Factor VII deficiency. Factor VII is the only factor in the extrinsic pathway alone, so it prolongs the PT while the aPTT, which does not depend on it, stays normal. Early vitamin K deficiency can give the same pattern.

#106 · Evaluation

A patient has an isolated prolonged aPTT. Name the pathways the aPTT screens and the factors it is most sensitive to.

Answer

The aPTT screens the intrinsic and common pathways: factors XII, XI, IX, VIII, X, V, II and fibrinogen, plus prekallikrein and HMWK. It is especially sensitive to deficiencies of factors VIII, IX and XI.

#107 · Evaluation

Why is the aPTT used to monitor unfractionated heparin, and what is the usual therapeutic target?

Answer

Heparin enhances antithrombin inhibition of thrombin and factor Xa, prolonging the aPTT. The target is usually 1.5 to 2.5 times the control mean, but each lab should set its range by calibrating against anti-Xa heparin levels.

#108 · Evaluation

A patient has a prolonged aPTT but no bleeding, and factor XII is very low. What is the likely clinical significance?

Answer

Factor XII, prekallikrein and HMWK deficiencies prolong the aPTT in vitro but cause no bleeding in patients. They are lab findings only, so do not delay surgery. Confirm with specific factor assays.

#109 · Evaluation

In a 1:1 mixing study, a prolonged aPTT corrects to normal. What does this pattern indicate?

Answer

Correction suggests a factor deficiency, because normal plasma supplies the missing factor. At least 50 percent factor level in the mix is enough to normalize the clotting time. An inhibitor would not correct.

#110 · Evaluation

How does the incubated mixing study separate a time-dependent factor VIII inhibitor from a lupus anticoagulant?

Answer

A factor VIII inhibitor is time and temperature dependent: the aPTT of the mix prolongs after 1 to 2 hours at 37 C. A lupus anticoagulant is immediate acting and does not strengthen with incubation.

#111 · Evaluation

Which factor assay result separates hemophilia A from hemophilia B in a male with a prolonged aPTT and normal PT?

Answer

Low factor VIII activity with normal factor IX means hemophilia A. Low factor IX with normal factor VIII means hemophilia B. Both are X-linked recessive. Normal von Willebrand antigen helps exclude VWD as the cause of a low factor VIII.

#112 · Evaluation

Why can von Willebrand disease give a prolonged aPTT, and which tests help diagnose it?

Answer

VWF carries and protects factor VIII, so low VWF can lower factor VIII and prolong the aPTT. Diagnosis uses VWF antigen, VWF activity (ristocetin cofactor or equivalent) and factor VIII. Platelet count is usually normal.

#113 · Evaluation

Why do factor assays use dilutions of patient plasma, and what does non-parallelism with the standard curve suggest?

Answer

Several dilutions are tested to check that results are parallel to the reference curve. Non-parallel results suggest an inhibitor or interference, such as a lupus anticoagulant, that is falsely lowering the factor activity.

#114 · Evaluation

What does a low fibrinogen with a prolonged thrombin time suggest, and what is a common non-disease cause?

Answer

Hypofibrinogenemia or dysfibrinogenemia can prolong the thrombin time. Heparin contamination is a common non-disease cause because heparin inhibits thrombin. A reptilase time stays normal if heparin is the cause.

#115 · Evaluation

Why is a negative D-dimer useful in excluding venous thromboembolism, and what limits a positive result?

Answer

D-dimer is a breakdown product of cross-linked fibrin, so a negative result in a low pretest probability patient makes VTE unlikely. A positive result is not specific: surgery, pregnancy, infection, cancer and age also raise it.

#116 · Evaluation

What is the typical laboratory pattern in acute disseminated intravascular coagulation (DIC)?

Answer

Prolonged PT and aPTT, low platelets, low fibrinogen, and markedly raised D-dimer. A falling platelet count and fibrinogen on repeat testing support the diagnosis. Fibrinogen can look normal early because it is an acute phase reactant.

#117 · Evaluation

Which blood film finding supports DIC, and why is trending results more informative than a single value?

Answer

Schistocytes (fragmented red cells) with thrombocytopenia support microangiopathic change. Trending platelets, fibrinogen, PT and D-dimer over hours shows direction, because DIC is dynamic and a single normal value may not rule it out.

#118 · Evaluation

Why should protein C and protein S testing not be done during acute thrombosis or warfarin therapy?

Answer

Both are vitamin K dependent and fall with warfarin, and acute clotting consumes them, so levels are falsely low. Test after the acute event, ideally off warfarin for 2 weeks or more, or interpret with caution.

#119 · Evaluation

Antithrombin levels fall during heparin therapy. Why does this matter when testing for antithrombin deficiency?

Answer

Heparin therapy and acute thrombosis can lower antithrombin, giving a falsely low result. Interpret with the clinical context and repeat the test later, when the patient is off heparin and not acutely ill.

#120 · Evaluation

What is the most common inherited cause of activated protein C resistance, and how is it confirmed?

Answer

Factor V Leiden, a point mutation (R506Q) that makes factor Va resistant to cleavage by activated protein C. It is confirmed by PCR genotyping. An APC resistance clotting assay is a functional screen.

#121 · Evaluation

How do direct oral anticoagulants (DOACs) such as rivaroxaban and dabigatran affect routine coagulation assays?

Answer

They can prolong the PT or aPTT depending on drug and reagent, but the effect is unreliable. Dabigatran (direct thrombin inhibitor) markedly prolongs thrombin time. Normal screening tests do not exclude drug presence.

#122 · Evaluation

When is an anti-Xa assay used, and what drug-specific calibration is required?

Answer

It measures activity of heparin (unfractionated or low molecular weight) or anti-Xa DOACs such as rivaroxaban and apixaban. The assay must be calibrated with the specific drug being measured, not a generic heparin calibrator.

#123 · Evaluation

A film shows schistocytes and low platelets, with prolonged PT and aPTT, low fibrinogen and high D-dimer. What is the likely picture and the first lab step?

Answer

This points to DIC. The first lab step is to notify the clinician promptly and repeat platelets, PT, aPTT, fibrinogen and D-dimer to follow trend, while checking for clotted or underfilled tube artifacts.

#124 · Evaluation

A CBC shows anemia and thrombocytopenia with schistocytes, yet PT, aPTT and fibrinogen are normal. What condition is suggested?

Answer

Thrombotic microangiopathy such as TTP or HUS. Normal coagulation tests help separate it from DIC. This is a hematologic emergency, so call the result to the clinician immediately.

#125 · Evaluation

What does a high RDW together with a low MCV most often suggest?

Answer

Iron deficiency: the red cells vary widely in size (anisocytosis) and are small on average. Thalassemia trait usually shows a low MCV with a normal or only mildly raised RDW, which helps separate the two.

#126 · Preparation and Testing

What spreader angle is used when making a wedge blood smear, and what happens if it changes?

Answer

Use about 30 to 45 degrees. A steeper angle gives a shorter, thicker film; a shallower angle gives a longer, thinner film. Adjust the angle to the patient's hematocrit.

#127 · Preparation and Testing

How big should the blood drop be for a wedge smear, and where is it placed on the slide?

Answer

A small drop, a few millimetres across, placed near one end of the slide. An oversized drop produces a thick smear that runs too far and has no usable feathered edge.

#128 · Preparation and Testing

How does spreading speed change the finished wedge smear?

Answer

A fast, smooth push gives a shorter, thicker film. A slow push gives a longer, thinner film. Hesitating or stopping mid-push causes ridges, so spread in one steady motion.

#129 · Preparation and Testing

Name common causes of a smear that is too thick.

Answer

A drop that is too large, a spreader angle that is too steep, a fast push, or a high hematocrit. Cells overlap and stain dark, so morphology cannot be read.

#130 · Preparation and Testing

Name common causes of a smear that is too thin.

Answer

A small drop, a shallow spreader angle, a slow push, or a low hematocrit (anemia). The film is long with few cells, and the feathered edge can run off the slide.

#131 · Preparation and Testing

What usually causes holes or streaks in a blood smear?

Answer

Holes come from grease or fingerprints on the slide. Streaks and ridges come from a dirty or chipped spreader, uneven pressure, or hesitation. Use clean, grease-free slides and a smooth spreader.

#132 · Preparation and Testing

What should a well-made wedge smear look like at its end?

Answer

It covers about two thirds of the slide, has visible side margins, and ends in a feathered edge, a smooth tongue shape. The film should not run off the end of the slide.

#133 · Preparation and Testing

What role does methanol play in Wright and Wright-Giemsa stains, and what buffer pH is typical?

Answer

The methanol in the stain fixes the cells to the slide. Buffer pH is typically about 6.8, which gives balanced red cell pink and white cell nuclear purple.

#134 · Preparation and Testing

Why must buffer pH be controlled during Romanowsky staining?

Answer

Staining depends on acidic and basic dyes binding differently at a given pH. A small shift changes how red cells, nuclei and granules colour, so use freshly checked buffer.

#135 · Preparation and Testing

A smear looks too blue with gray-blue red cells. What are likely causes?

Answer

Buffer or rinse water too alkaline, staining time too long, or a smear that is too thick. Red cells turn blue-gray and eosinophil granules look gray or blue instead of orange-red.

#136 · Preparation and Testing

A smear looks too pink with pale white cell nuclei. What are likely causes?

Answer

Buffer or rinse water too acidic, too short a staining time, or over-rinsing. Red cells look bright red and nuclei look pale, with eosinophil granules overly bright.

#137 · Preparation and Testing

What causes stain precipitate on a blood film, and how is it prevented?

Answer

Unfiltered or old stain, dirty slides, or stain allowed to dry on the slide. Filter the stain, keep it covered, and rinse without pouring off the stain first so debris floats away.

#138 · Preparation and Testing

What is the purpose of Kohler illumination?

Answer

It gives even, bright illumination with the best resolution and contrast. Focus the field diaphragm, centre the condenser, then set the aperture diaphragm for the objective in use.

#139 · Preparation and Testing

Why is oil used with the 100x objective, and what is the total magnification?

Answer

Immersion oil has a refractive index close to glass, so less light is lost and resolution improves. With a 10x eyepiece, total magnification is 1000x.

#140 · Preparation and Testing

Where on the smear is a manual differential performed, and how are fields moved?

Answer

In the thin monolayer area where red cells barely touch, just behind the feathered edge. Move in a battlement (crenellated) track so the same area is not counted twice.

#141 · Preparation and Testing

How do you correct a WBC count when nucleated red cells are present?

Answer

Corrected WBC = WBC x 100 / (100 + NRBC per 100 WBC). Example: 10.0 with 25 NRBC gives 10.0 x 100 / 125 = 8.0 x10^9/L.

#142 · Preparation and Testing

How is a platelet estimate made from a blood film, and what artifact should you check?

Answer

Count platelets in several oil immersion fields in the monolayer, average, and multiply by the lab's set factor. Check the feathered edge for platelet clumps, which falsely lower automated counts.

#143 · Preparation and Testing

Why compare a smear WBC estimate with the automated white cell count?

Answer

It is a quick quality check. Averaging white cells over several fields should roughly match the instrument count, and a clear mismatch prompts a recount or investigation of the specimen.

#144 · Preparation and Testing

What are the dimensions and volume of one large square of a Neubauer hemocytometer?

Answer

Each large square is 1 mm x 1 mm with a chamber depth of 0.1 mm, so its volume is 0.1 mm3, which equals 0.1 microlitre.

#145 · Preparation and Testing

Which Neubauer areas are counted for WBC and for RBC?

Answer

WBC: the four corner large squares. RBC: five small squares in the central large square (four corners and the centre). Platelets are counted in the central square, ideally with phase contrast.

#146 · Preparation and Testing

Which cells touching a square's borders are counted in a hemocytometer?

Answer

Count cells touching the top and left borders, and skip cells touching the bottom and right borders. This stops cells on shared lines being counted twice.

#147 · Preparation and Testing

Worked WBC example: 80 cells in 4 large squares, 1:20 dilution. What is the count?

Answer

Volume counted is 4 x 0.1 = 0.4 microlitre. Count = 80 x 20 / 0.4 = 4000 per microlitre, which is 4.0 x10^9/L.

#148 · Preparation and Testing

Worked RBC example: 400 cells in 5 small squares, 1:200 dilution. What is the count?

Answer

Five small squares are 5/25 mm2 = 0.2 mm2, times 0.1 mm depth = 0.02 microlitre. Count = 400 x 200 / 0.02 = 4.0 x10^12/L.

#149 · Preparation and Testing

What does new methylene blue stain in a reticulocyte, and how is the specimen prepared?

Answer

It precipitates residual RNA as a blue network or granules in young red cells. Mix equal parts blood and stain, incubate roughly 10 to 15 minutes, then make a smear.

#150 · Preparation and Testing

How does a Miller disc speed up a reticulocyte count?

Answer

The disc has a large square B and a small square A (one ninth the area). Count retics in B and red cells in A: retic % = retics in B / (RBC in A x 9) x 100.

#151 · Preparation and Testing

What physical principle does the Coulter method use to count and size blood cells?

Answer

Cells are drawn through a small aperture between two electrodes. Each cell displaces conductive diluent and causes a voltage pulse. Pulse count equals cell number, and pulse height is proportional to cell volume.

#152 · Preparation and Testing

On an impedance analyzer, why do two cells passing through the aperture together cause a counting error?

Answer

Coincidence passage is recorded as one larger pulse, so the count is falsely low and the size falsely high. Analyzers apply a coincidence correction, and heavily elevated counts may need dilution.

#153 · Preparation and Testing

In impedance counting, how is the WBC count obtained when the RBCs are present in the same sample?

Answer

A lytic reagent destroys the RBCs and strips WBC cytoplasm to a small nuclear remnant, so only WBC nuclei are counted in the WBC channel. The lysed sample also supplies the hemoglobin measurement.

#154 · Preparation and Testing

What cell property does forward-angle light scatter in a flow cell mainly measure?

Answer

Forward scatter reflects cell size and volume. Side scatter reflects internal complexity, such as granules and nuclear lobulation. Plotting the two together separates lymphocytes, monocytes and granulocytes.

#155 · Preparation and Testing

How do fluorescent nucleic acid dyes help an analyzer count reticulocytes or nucleated red cells?

Answer

The dye binds RNA or DNA, and fluorescence intensity reflects nucleic acid content. Reticulocytes carry residual RNA and nRBCs carry DNA, so each group appears in its own region on the scattergram.

#156 · Preparation and Testing

What is a major advantage of a flow-cytometric (laser-based) WBC differential over a manual 100-cell count?

Answer

Thousands of cells are classified, giving much better statistical precision. Manual counts of 100 cells carry a large sampling error, especially for low-frequency cell types such as basophils.

#157 · Preparation and Testing

Name the reagent chemistry behind the cyanmethemoglobin method and its measured wavelength.

Answer

Ferricyanide oxidizes hemoglobin to methemoglobin, and cyanide converts it to stable cyanmethemoglobin, read at 540 nm. Absorbance is proportional to hemoglobin concentration, and the method is the international reference.

#158 · Preparation and Testing

Why do many analyzers use sodium lauryl sulfate (SLS) hemoglobin instead of cyanide?

Answer

SLS-hemoglobin avoids toxic cyanide waste. It forms a stable SLS-methemoglobin complex measured by absorbance, and it correlates well with the cyanmethemoglobin reference method.

#159 · Preparation and Testing

Which sample conditions falsely raise a spectrophotometric hemoglobin result?

Answer

Lipemia, very high WBC counts and heavy turbidity scatter light and raise absorbance. Correct by plasma replacement (spin and blank the plasma) for lipemia, or by centrifuging the lysate for a high WBC.

#160 · Preparation and Testing

What is the usual centrifugation setting for a microhematocrit and why must the reading skip the buffy coat?

Answer

Spin about 5 minutes at roughly 10,000 to 12,000 x g. Read only the red cell column, since the buffy coat holds WBCs and platelets and is not part of the packed red cell volume.

#161 · Preparation and Testing

What is trapped plasma in a microhematocrit, and which direction does it push the result?

Answer

A small plasma volume (about 1 to 3 percent) stays between packed red cells, so the spun hematocrit reads slightly high. It increases with spherocytes, sickle cells, macrocytes and high hematocrits.

#162 · Preparation and Testing

How does an impedance analyzer produce its hematocrit value?

Answer

It calculates hematocrit as RBC count multiplied by mean cell volume (MCV), divided by 10 for percent. No spinning is involved, so trapped plasma does not apply to the calculated value.

#163 · Preparation and Testing

Give the formula for MCV and a worked example with Hct 0.40 L/L and RBC 4.0 x10^12/L.

Answer

MCV (fL) = Hct (L/L) x 1000 divided by RBC (x10^12/L). Here 0.40 x 1000 = 400, then 400 divided by 4.0 gives 100 fL, which is at the upper edge of normal.

#164 · Preparation and Testing

Give the formula for MCH and calculate it for Hb 120 g/L and RBC 4.0 x10^12/L.

Answer

MCH (pg) = Hb (g/L) divided by RBC (x10^12/L). Here 120 divided by 4.0 equals 30 pg. MCH is the average hemoglobin mass in each red cell.

#165 · Preparation and Testing

Give the formula for MCHC and calculate it for Hb 120 g/L and Hct 0.36 L/L.

Answer

MCHC (g/L) = Hb (g/L) divided by Hct (L/L). Here 120 divided by 0.36 equals 333 g/L. It is the average hemoglobin concentration within packed red cells.

#166 · Preparation and Testing

Which red cell index is the most useful quick check for a wrong or spurious CBC, and what limit applies?

Answer

MCHC. A value above about 360 g/L is physiologically implausible and points to a falsely high hemoglobin (lipemia, hemolysis) or a falsely low hematocrit (cold agglutinins, RBC clumps).

#167 · Preparation and Testing

What does a histogram with two RBC populations (a bimodal curve) suggest?

Answer

Two distinct cell sizes are present, such as a dimorphic anemia, recent transfusion or a mix of microcytes and normal cells. Check the RDW and the blood film to confirm.

#168 · Preparation and Testing

What does a platelet histogram with a long tail running into the RBC region suggest?

Answer

Giant platelets or platelet clumps, or small RBC fragments, are overlapping the size threshold. Review the film and the flags before reporting, since the count may be falsely low or high.

#169 · Preparation and Testing

What interference do cold agglutinins cause on a CBC, and how is it resolved?

Answer

RBCs clump in the cold, giving a falsely low RBC count and Hct, a high MCV and a very high MCHC. Warm the sample to 37 C and rerun, or use warmed saline replacement.

#170 · Preparation and Testing

How do nucleated red blood cells (nRBCs) interfere with the automated WBC count?

Answer

nRBC nuclei resist lysis and are counted as WBCs, falsely raising the WBC count. Many analyzers flag and correct this, or the tech corrects manually: corrected WBC = WBC x 100 divided by (nRBC per 100 WBC + 100).

#171 · Preparation and Testing

Which CBC result is a classic problem caused by EDTA-dependent platelet clumping, and what is the next step?

Answer

A falsely low platelet count with clumps on the film. Recollect in sodium citrate (multiply the result by 1.1 for dilution) or use a fingerstick sample, then rerun and review the film.

#172 · Preparation and Testing

What is a delta check on a CBC and what does it detect?

Answer

A comparison of the current result with the patient's previous result. A change beyond a set limit, such as MCV shifting by several fL, can flag a mislabelled specimen or a wrong-patient sample.

#173 · Preparation and Testing

State the rule of three for CBC results and what a failure means.

Answer

Hb (g/L) is about 3 x RBC (x10^12/L) x 10, and Hct (%) is about 3 x Hb (g/dL). When the numbers do not fit, suspect interference or an error and investigate before reporting.

#174 · Preparation and Testing

In the Westergren ESR method, what tube length, anticoagulant ratio and read time are used?

Answer

A 200 mm vertical tube filled with 4 parts blood to 1 part sodium citrate, read as the millimetres of red cell fall after exactly 1 hour. The tube must stand at room temperature, away from vibration or tilt.

#175 · Preparation and Testing

Name factors that falsely raise or lower the ESR.

Answer

Raised by anemia, high fibrinogen and immunoglobulins, and a tilted tube. Lowered by polycythemia, sickle cells, spherocytes, microcytosis and clotted samples. Delayed testing (over 4 hours) also lowers the ESR.

#176 · Preparation and Testing

What does the sickle solubility test detect, and what are its main limits?

Answer

Hemoglobin S insolubility in reduced phosphate buffer gives a turbid solution. It cannot tell trait from disease, and false negatives occur in very low Hb S levels or in infants under 6 months. Confirm with electrophoresis or HPLC.

#177 · Preparation and Testing

At what pH is routine alkaline hemoglobin electrophoresis run, and in which direction do hemoglobins migrate?

Answer

Alkaline electrophoresis runs at pH 8.4 to 8.6. Hemoglobin is negatively charged at this pH, so it migrates toward the anode (positive pole). Order from anode: A, then F, S, and C nearest the origin.

#178 · Preparation and Testing

Why is acid (citrate agar, pH about 6.0) electrophoresis used after an alkaline run shows a band in the S position?

Answer

Acid agar separates Hb S from Hb D and Hb G, which co-migrate with S on alkaline media. In acid agar, S moves apart from D and G, which move with A. It also separates C from E and O-Arab.

#179 · Preparation and Testing

What does a raised Hb A2 on HPLC (above about 3.5 percent) suggest, and what is the typical companion finding?

Answer

A raised Hb A2 suggests beta thalassemia trait. The CBC usually shows microcytosis (low MCV) with a normal or raised red cell count. Iron deficiency can lower A2 and mask the trait.

#180 · Preparation and Testing

In which condition is red cell osmotic fragility increased, and what does the cell shape explain?

Answer

Hereditary spherocytosis. Spherocytes have a low surface-area-to-volume ratio, so they swell and lyse in less dilute saline than normal discs. Thalassemia and iron deficiency show decreased fragility (target cells).

#181 · Preparation and Testing

What is the principle of the Kleihauer-Betke test, and how do fetal and adult cells look?

Answer

Acid elution of a blood smear. Hb F resists acid, so fetal cells stain bright pink-red, while adult cells lose their hemoglobin and appear as pale ghosts. Counting quantifies the fetomaternal bleed.

#182 · Preparation and Testing

Which stain detects reticulocytes, and what does it make visible?

Answer

New methylene blue, a supravital stain used on unfixed living cells. It precipitates residual ribosomal RNA in young red cells as a blue reticulum or granules. Count 1000 red cells for the percentage.

#183 · Preparation and Testing

How are Heinz bodies detected, and what are they made of?

Answer

Heinz bodies are denatured, precipitated hemoglobin seen with a supravital stain such as crystal violet or methyl violet, not on Wright stain. They appear as purple inclusions at the cell edge, often in G6PD deficiency.

#184 · Preparation and Testing

Which cytochemical stains are positive in myeloblasts but negative in lymphoblasts?

Answer

Myeloperoxidase and Sudan black B. Both mark myeloid granules (Auer rods too). Lymphoblasts are negative, so these stains help separate AML from ALL when morphology is unclear.

#185 · Preparation and Testing

Which esterase stain marks monocytic cells, and what inhibits it?

Answer

Nonspecific esterase (alpha-naphthyl butyrate or acetate) is positive in monocytic cells and is inhibited by sodium fluoride. Chloroacetate esterase is the specific esterase and marks neutrophil lineage.

#186 · Preparation and Testing

Which stain is positive in hairy cell leukemia even after tartrate is added?

Answer

TRAP, tartrate-resistant acid phosphatase. Hairy cells keep acid phosphatase activity in the presence of tartrate, whereas most other cells lose it. Flow cytometry (CD11c, CD25, CD103) is now often preferred.

#187 · Preparation and Testing

How does leukocyte alkaline phosphatase (LAP) score help separate CML from a leukemoid reaction?

Answer

LAP is low in chronic myeloid leukemia and high in a leukemoid reaction or infection. It is largely replaced by BCR::ABL1 molecular testing, but the pattern is still a classic recall point.

#188 · Preparation and Testing

On flow cytometry, what do forward scatter and side scatter measure?

Answer

Forward scatter reflects cell size. Side scatter reflects internal complexity or granularity. Together with CD45 intensity, they separate lymphocytes, monocytes, granulocytes and blasts on the plot.

#189 · Preparation and Testing

Which CD markers identify T cells, B cells and immature blasts?

Answer

CD3 marks T cells and CD19 (or CD20) marks B cells. CD34 and low CD45 mark immature blasts, and TdT marks immature lymphoid cells. CD14 marks monocytes, and CD138 marks plasma cells.

#190 · Preparation and Testing

Why is CD45 versus side scatter a standard first gating plot in leukemia immunophenotyping?

Answer

Blasts show dim CD45 with low side scatter, lying apart from mature lymphocytes (bright CD45) and granulocytes (high side scatter). It lets the analyst gate the blast population and assign lineage.

#191 · Preparation and Testing

Why is the first small volume of a bone marrow aspirate used for smears?

Answer

The first 1 to 2 mL is least diluted with peripheral blood, so the smears show true marrow cellularity and morphology. Later pulls are hemodiluted. Smears are made at the bedside before the sample clots.

#192 · Preparation and Testing

How is a bone marrow trephine core biopsy processed before staining?

Answer

It is fixed, then decalcified to soften the bone, embedded in paraffin and sectioned for H and E. Touch preparations are made from the fresh core before fixation. Over-decalcifying damages nuclear detail.

#193 · Preparation and Testing

What is the correct blood to anticoagulant ratio for coagulation specimens, and which anticoagulant is used?

Answer

Nine parts blood to one part 3.2 percent (0.109 mol/L) buffered sodium citrate in a light blue tube. Underfilling causes excess citrate, which dilutes the sample and falsely prolongs clotting times.

#194 · Preparation and Testing

What must be done when the patient hematocrit is above 55 percent for coagulation testing?

Answer

Reduce the citrate volume in the tube. A high hematocrit leaves less plasma, so the fixed citrate is relatively excessive and falsely prolongs the PT and aPTT. Use the formula supported by the lab policy.

#195 · Preparation and Testing

What defines platelet-poor plasma for coagulation testing, and why does it matter?

Answer

Plasma with a platelet count below 10 x10^9/L, made by centrifuging the citrate tube. Residual platelets release phospholipid and can shorten clotting times, notably masking a lupus anticoagulant or heparin effect.

#196 · Preparation and Testing

How is the INR calculated from the patient PT?

Answer

INR = (patient PT divided by the mean normal PT) raised to the power of the ISI. A lower ISI, closer to 1.0, means a more sensitive thromboplastin calibrated to the WHO reference.

#197 · Preparation and Testing

What is the mean normal prothrombin time (MNPT) and how is it established?

Answer

The geometric mean of PT results from at least 20 healthy, untreated adults, for each new thromboplastin lot and instrument. It is the denominator in the INR calculation.

#198 · Preparation and Testing

What are the activators in aPTT reagents, and what do they trigger?

Answer

Silica, kaolin, celite or ellagic acid activate the contact factors (XII, XI) in the presence of partial thromboplastin phospholipid and calcium. The aPTT is prolonged by deficiencies of VIII, IX, XI and XII.

#199 · Preparation and Testing

Why can different aPTT reagents give different heparin therapeutic ranges?

Answer

Reagents differ in activator type and phospholipid content, so sensitivity to heparin varies. Each laboratory must verify its therapeutic aPTT range for every new reagent lot against anti-Xa heparin levels.

#200 · Preparation and Testing

What is the principle of the Clauss method for fibrinogen, and what is a typical reference range?

Answer

Diluted plasma (usually 1 in 10) is clotted with excess thrombin, and the clotting time is inversely proportional to fibrinogen, read from a calibration curve. Normal is about 2.0 to 4.0 g/L.

#201 · Preparation and Testing

Which test is prolonged by heparin: thrombin time or reptilase time?

Answer

Thrombin time is prolonged by heparin. Reptilase time is not, because reptilase is unaffected by heparin. Both are prolonged by low fibrinogen, dysfibrinogenemia and fibrin degradation products.

#202 · Preparation and Testing

What is the main clinical use of a D-dimer result in suspected venous thromboembolism?

Answer

A negative D-dimer, in a patient with low or moderate pretest probability, helps rule out DVT or PE. It has high negative predictive value but poor specificity, so a positive result does not confirm thrombosis.

#203 · Preparation and Testing

Which fibrin degradation product does a D-dimer assay detect, and why does it differ from a plain FDP test?

Answer

D-dimer detects fragments from cross-linked fibrin broken down by plasmin, so it signals fibrinolysis of a formed clot. A plain FDP test also reacts to fibrinogen breakdown products.

#204 · Preparation and Testing

Name three common causes of a falsely elevated or non-specific positive D-dimer.

Answer

Pregnancy, recent surgery or trauma, malignancy, infection, advanced age, and liver disease all raise D-dimer without thrombosis. Rheumatoid factor can also interfere in some latex methods.

#205 · Preparation and Testing

In a 1:1 mixing study, what does correction of a prolonged aPTT suggest?

Answer

Correction into the reference interval suggests a factor deficiency, because the normal plasma supplies the missing factor. Failure to correct suggests an inhibitor such as a lupus anticoagulant or a factor antibody.

#206 · Preparation and Testing

Why is an incubated mix (37 C for 1 to 2 hours) performed in a mixing study?

Answer

Some factor inhibitors, such as factor VIII antibodies, act slowly and time-dependently. Immediate mixing may correct, but the incubated mix will prolong again, revealing the inhibitor.

#207 · Preparation and Testing

What is the principle of a one-stage factor VIII assay?

Answer

Patient plasma is diluted and added to factor VIII deficient plasma, then an aPTT is run. The clotting time is read against a reference curve to give percent activity.

#208 · Preparation and Testing

Why are factor assays run at several dilutions, and what does non-parallelism suggest?

Answer

Multiple dilutions test parallelism with the reference curve. Non-parallel lines suggest an inhibitor or interfering substance, and the factor activity should not be reported as a true value.

#209 · Preparation and Testing

What is the screening principle of the dilute Russell viper venom time (dRVVT)?

Answer

Russell viper venom directly activates factor X in the presence of phospholipid and calcium. Low phospholipid concentration makes the test sensitive to a lupus anticoagulant, which prolongs the clotting time.

#210 · Preparation and Testing

How is a lupus anticoagulant confirmed after a prolonged dRVVT screen?

Answer

Repeat the dRVVT with a high phospholipid concentration. Shortening of the clotting time by a set percentage or ratio shows the inhibitor was phospholipid dependent, confirming a lupus anticoagulant.

#211 · Preparation and Testing

Why should lupus anticoagulant testing be done on double-centrifuged, platelet-poor plasma?

Answer

Residual platelets release phospholipid that can neutralize the antibody and cause a false negative. Platelet-poor plasma (platelets under about 10 x10^9/L) is required.

#212 · Preparation and Testing

Which anticoagulant therapies can cause a false positive lupus anticoagulant result?

Answer

Unfractionated heparin (when not neutralized), direct thrombin inhibitors, and direct oral anticoagulants such as rivaroxaban or apixaban can prolong dRVVT, giving false positives. Check the patient's medications.

#213 · Preparation and Testing

What is the principle of a chromogenic anti-Xa assay for heparin monitoring?

Answer

Heparin and antithrombin inhibit a fixed excess of factor Xa. Residual Xa cleaves a chromogenic substrate, and the colour intensity is inversely proportional to heparin concentration.

#214 · Preparation and Testing

Why are chromogenic assays less affected by a lupus anticoagulant than clot-based tests?

Answer

They measure colour from substrate cleavage, not clot formation, so phospholipid-dependent antibodies do not interfere. This makes them useful for monitoring heparin in patients with a baseline prolonged aPTT.

#215 · Preparation and Testing

How does a chromogenic antithrombin activity assay work?

Answer

Patient antithrombin, with excess heparin, inhibits added thrombin or factor Xa. The remaining enzyme cleaves a chromogenic substrate, so colour is inversely proportional to antithrombin activity.

#216 · Preparation and Testing

Which platelet agonists are commonly used in light transmission aggregometry?

Answer

ADP, collagen, epinephrine, arachidonic acid, and ristocetin. Each probes a different pathway, so the pattern of response helps identify the platelet defect.

#217 · Preparation and Testing

How is platelet aggregation measured in light transmission aggregometry?

Answer

Platelet-rich plasma starts turbid. As platelets aggregate after an agonist is added, the sample clears and light transmission rises, measured against platelet-poor plasma as 100 percent.

#218 · Preparation and Testing

What does a PFA-100 closure time measure, and what is a common cause of a prolonged result?

Answer

It measures time for a platelet plug to close an aperture under high shear flow. Prolonged results occur with aspirin, von Willebrand disease, severe thrombocytopenia, and a low hematocrit.

#219 · Preparation and Testing

Name patient or specimen factors that cause false results in platelet function testing.

Answer

Recent aspirin or NSAID use, a low platelet count, a delayed or cold specimen, and a traumatic draw. Specimens should be kept at room temperature, never refrigerated, and tested within a few hours.

#220 · Preparation and Testing

Describe the difference between optical and mechanical clot detection on coagulation analyzers.

Answer

Optical methods detect a change in light transmission or scatter as fibrin forms. Mechanical methods detect increased viscosity, such as a steel ball movement slowing, when the clot forms.

#221 · Preparation and Testing

Which specimen interferences affect optical clot detection, and which method is less affected?

Answer

Lipemia, icterus, and hemolysis alter turbidity or absorbance and can interfere with optical detection. Mechanical detection is largely unaffected, so it is often used for such samples.

#222 · Preparation and Testing

What is the usual diluent and counting chamber for a CSF total cell count?

Answer

CSF is usually counted undiluted in a hemocytometer. Dilute with isotonic saline if cell numbers are high, and use dilute acetic acid only when lysing red cells is needed to count nucleated cells.

#223 · Preparation and Testing

Why must hyaluronidase or a saline diluent be used for synovial fluid counts instead of acetic acid?

Answer

Acetic acid causes mucin (hyaluronate) clot formation, trapping cells and spoiling the count. Use saline, or pretreat viscous fluid with hyaluronidase to reduce viscosity.

#224 · Preparation and Testing

How is a cytocentrifuge slide made from a body fluid, and what is a common artifact?

Answer

A small fluid volume is spun onto a slide in a cytospin funnel, concentrating cells in a spot. Cells may shrink or distort, and the stain is often a Wright-Giemsa stain.

#225 · Preparation and Testing

What is the advantage of phase-contrast microscopy for manual platelet counts?

Answer

Phase contrast makes unstained platelets stand out as bright, refractile bodies against a dark background, which makes them easier to tell apart from debris. Count them in a hemocytometer after diluting with ammonium oxalate.

#226 · Preparation and Testing

How long can a citrated coagulation specimen for PT and aPTT be held at room temperature before testing?

Answer

PT is generally stable up to 24 hours as a capped whole blood or separated plasma, while aPTT should be tested within 4 hours if unfractionated heparin is monitored. Follow the lab's validated limits.

#227 · Specimen Collection

What is the CLSI order of draw for the common venipuncture tubes?

Answer

Blood culture, light blue citrate, red or gold serum (with or without gel), green heparin, lavender EDTA, then grey fluoride oxalate. Hook: the additive tubes come after the ones that must stay clean.

#228 · Specimen Collection

Why must the citrate tube be drawn before the EDTA tube?

Answer

EDTA carried over on the needle chelates calcium and potassium and contaminates the citrate tube, falsely prolonging PT and aPTT because coagulation tests depend on calcium being restored.

#229 · Specimen Collection

Which additive in the lavender tube makes it the tube of choice for a CBC?

Answer

K2EDTA or K3EDTA chelates calcium, which prevents clotting while preserving cell size and morphology. It works by binding calcium, so it cannot be used for coagulation testing.

#230 · Specimen Collection

What is the function of heparin tubes, and which test group do they suit?

Answer

Heparin (lithium or sodium) activates antithrombin to block thrombin, giving plasma for chemistry. Sodium heparin is wrong for sodium testing, and heparin is unsuitable for blood films because it distorts stain background.

#231 · Specimen Collection

What does sodium fluoride in a grey top tube do for glucose testing?

Answer

Fluoride inhibits glycolysis (enolase), so glucose in the tube stays stable for hours instead of falling about 5 to 7 percent per hour from cell metabolism. Oxalate is the anticoagulant.

#232 · Specimen Collection

How many times should an EDTA tube be gently inverted right after collection?

Answer

Invert 8 to 10 times, gently, to mix EDTA fully with the blood. Under-mixing allows microclots and platelet clumps, while vigorous shaking causes hemolysis.

#233 · Specimen Collection

What does a microclot in an EDTA tube do to CBC results?

Answer

Clots or fibrin strands trap cells, giving falsely low platelet and sometimes WBC and RBC counts. Examine for clots with applicator sticks and recollect if present.

#234 · Specimen Collection

What happens to blood cell morphology when an EDTA specimen sits for many hours before a film is made?

Answer

Storage artifact appears: neutrophils swell, lobes degenerate, vacuoles form, and RBCs may crenate or become spheroidal. Make films within 2 to 3 hours for reliable morphology.

#235 · Specimen Collection

What is the required blood to anticoagulant ratio for a sodium citrate coagulation tube?

Answer

Nine parts blood to one part 3.2 percent sodium citrate (9:1). The tube must be filled to its marked line, because the citrate volume is fixed for a full draw.

#236 · Specimen Collection

How does a short draw affect a citrate coagulation specimen?

Answer

Too much citrate relative to plasma over-dilutes the sample and binds extra calcium, falsely prolonging PT and aPTT. Most labs reject tubes filled below 90 percent.

#237 · Specimen Collection

When must citrate volume be adjusted for coagulation tubes, and in which direction?

Answer

When the hematocrit is above 55 percent, plasma volume is reduced, so excess citrate must be lowered by adding less citrate for the smaller plasma volume. Otherwise PT and aPTT are falsely prolonged.

#238 · Specimen Collection

How should a citrate tube be mixed after collection, and why not shake it?

Answer

Gently invert 3 to 4 times. Vigorous shaking activates platelets and factors and can cause hemolysis, which affects clotting assays, so avoid it.

#239 · Specimen Collection

What is the first step of patient identification before a bedside collection?

Answer

Ask an outpatient to state their full name and a second identifier such as date of birth, and compare with the requisition. Never prompt with the answer. For inpatients, also check the wristband.

#240 · Specimen Collection

When must a tube be labelled during collection?

Answer

Label in the patient's presence, immediately after the draw, with two identifiers, date, time and collector initials. Never pre-label tubes or label away from the patient.

#241 · Specimen Collection

What should you do if an unlabelled or mislabelled specimen reaches the lab?

Answer

Reject it and recollect, since it cannot be reliably tied to the patient. For irreplaceable specimens, follow lab policy with documented clinician and collector verification.

#242 · Specimen Collection

Why does a patient lying down versus sitting upright change lab results?

Answer

Moving from supine to upright shifts plasma water into tissue, raising hemoglobin, hematocrit, protein and cell counts by about 5 to 10 percent. Standardize posture for serial results.

#243 · Specimen Collection

When should trough levels for therapeutic drug monitoring be collected?

Answer

Collect the trough just before the next scheduled dose, when the drug level is lowest. Collecting after the dose gives a falsely high value.

#244 · Specimen Collection

How long can a tourniquet stay on before hemoconcentration affects results?

Answer

Keep it on no longer than 1 minute. Longer application causes hemoconcentration and raises protein, calcium, cholesterol and cell counts, and can affect potassium.

#245 · Specimen Collection

Which vein is the first choice for adult routine venipuncture?

Answer

The median cubital vein of the antecubital fossa, because it is large, well anchored and away from major nerves and arteries. Basilic is the last choice due to the nearby brachial artery and median nerve.

#246 · Specimen Collection

What is the correct way to wipe the first drop in a fingerstick for CBC?

Answer

Wipe away the first drop, which has tissue fluid contamination, then collect the free flowing following drops. Do not squeeze hard, since tissue fluid dilutes the blood and lowers counts.

#247 · Specimen Collection

In what order should capillary tubes be filled to limit platelet clumping?

Answer

Fill the lavender EDTA microtainer first, since platelets clump rapidly at the puncture site, then other additive tubes, then serum last. Mix EDTA tubes promptly.

#248 · Specimen Collection

Which site is used for capillary puncture in newborns and infants under one year?

Answer

The lateral or medial plantar surface of the heel. Avoid the central heel and the posterior curve, where the calcaneus is close and can be injured, causing osteomyelitis.

#249 · Specimen Collection

What limits blood volume drawn from infants, and what risk does overcollection carry?

Answer

Total blood volume is small, about 80 mL per kg in a newborn, so frequent draws can cause iatrogenic anemia. Follow lab limits, usually no more than about 3 percent of volume at once.

#250 · Specimen Collection

Why is blood not drawn from an arm with an IV infusing, and what is the alternative?

Answer

IV fluid dilutes the specimen and can contaminate it with the infusate. Use the opposite arm; if impossible, draw below the IV site after stopping the infusion for at least 2 minutes with nurse approval.

#251 · Specimen Collection

Name three collection errors that commonly cause in vitro hemolysis of a hematology specimen.

Answer

Small-gauge needle, vigorous tube shaking instead of gentle inversion, forcing blood through a syringe, drawing from a hematoma, or venipuncture before alcohol has dried. Prevent by good technique.

#252 · Specimen Collection

How does in vitro hemolysis change the CBC red cell results?

Answer

RBC count and hematocrit fall because cells are lysed, while measured hemoglobin stays true, so MCHC rises falsely. A high MCHC with a pink plasma is the clue.

#253 · Specimen Collection

Why do many labs reject grossly hemolyzed citrate tubes for coagulation testing?

Answer

Free hemoglobin interferes with optical clot detection, and the cell lysis suggests platelet and tissue factor activation. Results may be unreliable, so recollect by a clean venipuncture.

#254 · Specimen Collection

A CBC tube arrives with visible clots. What is the correct action and why?

Answer

Reject and request recollection. Clots trap cells and platelets, giving falsely low counts and unreliable indices. Never remove a clot and run the remaining blood.

#255 · Specimen Collection

What fill level makes a light blue citrate tube unacceptable for PT and aPTT?

Answer

Under about 90 percent of the draw volume. The 9:1 blood to citrate ratio is lost, excess citrate binds calcium, and clotting times are falsely prolonged.

#256 · Specimen Collection

A tube is past its expiry date. Why is it rejected for testing?

Answer

Vacuum may be lost and the additive may have degraded, so the draw volume and anticoagulant effect are unreliable. Discard the tube and recollect into an in-date tube.

#257 · Specimen Collection

Which temperature rule applies to citrate tubes for PT and aPTT during transport?

Answer

Keep at room temperature, do not refrigerate. Cold activates platelets and factor VII, which can shorten clotting times, so unspun capped samples are not placed on ice.

#258 · Specimen Collection

How long is an EDTA specimen for ESR generally stable at room temperature and at 4 C?

Answer

About 4 hours at room temperature and up to 24 hours at 4 C (ICSH and CLSI guidance). Bring refrigerated samples to room temperature before running.

#259 · Specimen Collection

Why should transport avoid rough agitation, such as dropping tubes or long pneumatic tube trips without validation?

Answer

Agitation causes in vitro hemolysis and can activate platelets. Pneumatic tube systems should be validated by the lab for the tube types and tests sent.

#260 · Specimen Collection

What is the usual stability limit of EDTA whole blood for a CBC, and what degrades first?

Answer

Most analyzers accept room temperature samples within about 24 hours. Platelet and white cell morphology, and indices like MCV, degrade first as cells swell.

#261 · Specimen Collection

Within what time should a blood film be made from an EDTA sample for best morphology?

Answer

Ideally within 2 to 3 hours of collection. Prolonged EDTA exposure causes artifacts such as crenated red cells, vacuolated neutrophils and degenerating lymphocytes.

#262 · Specimen Collection

For routine coagulation testing, how long is an unspun capped citrate sample acceptable for PT and for aPTT?

Answer

PT up to 24 hours at room temperature. aPTT is more labile: process within 4 hours, and within 1 hour if monitoring unfractionated heparin (CLSI H21).

#263 · Specimen Collection

What should you check on a requisition before collecting or testing a specimen?

Answer

Two patient identifiers, the ordering provider, test requested, collection date and time, and clinical information. Query any missing or illegible item before proceeding.

#264 · Specimen Collection

A CBC is ordered again two hours after a normal result on a stable outpatient. What is the best response?

Answer

Recognize a possible duplicate in the lab information system and contact the ordering provider per lab policy before collecting, to avoid unnecessary phlebotomy and cost.

#265 · Specimen Collection

What clue on a CBC suggests wrong blood in tube?

Answer

A delta check failure: hemoglobin, MCV or other indices differ sharply from the patient's previous result without a clinical reason such as transfusion or bleeding.

#266 · Specimen Collection

A suspected wrong blood in tube result has been flagged. What is the first action?

Answer

Hold the result and do not report it. Notify the supervisor, recollect with positive patient identification, compare results, and document an occurrence report per policy.

#267 · Specimen Collection

What does chain of custody mean for a legal specimen?

Answer

A documented, unbroken record of every person who handled the specimen, with dates, times and transfers, so the sample's identity and integrity can be proven in court.

#268 · Specimen Collection

What controls are used on a legal specimen such as one for forensic or paternity testing?

Answer

Verified patient identification, witnessed collection, tamper-evident seals, signed custody forms at each transfer, and secure storage. Follow the lab's legal specimen policy exactly.

#269 · Specimen Collection

When should an INR be drawn for a patient on warfarin?

Answer

At a consistent time of day, ideally before the next warfarin dose. The INR reflects the stable effect of the drug, so timing consistency helps trend comparison.

#270 · Specimen Collection

When is the peak anti-Xa level drawn for a patient on subcutaneous low molecular weight heparin?

Answer

About 4 hours after the dose, after at least the third or fourth dose has reached steady state. A trough is drawn just before the next dose when ordered.

#271 · Specimen Collection

What discard volume is generally used when drawing coagulation samples from a central line or catheter?

Answer

At least 5 mL or six times the dead space volume of the line, per lab policy. Stop infusions first and use a syringe or discard tube, never the flush solution.

#272 · Specimen Collection

Why is drawing from the arm opposite an IV infusion preferred?

Answer

It avoids dilution and contamination of the sample by infusing fluids or heparin. If a line must be used, pause infusions for at least 2 minutes and follow discard rules.

#273 · Specimen Collection

A patient has an unexpectedly prolonged aPTT and thrombin time but a normal PT, and a heparinized line was used. What is suspected, and what is done?

Answer

Heparin contamination of the sample. Report nothing as final, then recollect by clean venipuncture from the opposite arm, or after proper line discard, and rerun.

#274 · Specimen Collection

How should a blood sample for cold agglutinin testing be kept from collection until the serum or plasma is separated?

Answer

Keep it warm at about 37 C (warm pack or heat block) until separated. Cooling lets the antibody bind red cells, so cells clump and the antibody is lost into the clot.

#275 · Specimen Collection

Why can a cold agglutinin sample give a falsely high MCV and falsely low red cell count on an analyzer?

Answer

Red cells agglutinate in the cooled sample, so clumps are counted as single large cells. Warm the sample to 37 C and rerun, or use a prewarmed collection.

#276 · Specimen Collection

Why must a cryoglobulin specimen be collected in prewarmed tubes and transported at 37 C?

Answer

Cryoglobulins precipitate or gel below body temperature. Cooling before clot retraction traps them in the clot and gives a false negative result.

#277 · Specimen Collection

What is the usual tube type and handling for a cryoglobulin specimen?

Answer

Plain red-top (no gel, no anticoagulant) kept at 37 C to clot, then spun warm. The serum is then refrigerated and checked over several days for a cryoprecipitate.

#278 · Specimen Collection

Which anticoagulant is usually used for a bone marrow aspirate sent for morphology and flow cytometry?

Answer

EDTA (lavender) for morphology and counts. Flow cytometry often uses sodium heparin (green). Follow the local requisition, since cytogenetics also needs heparin.

#279 · Specimen Collection

Why should bone marrow aspirate smears be made at the bedside immediately after collection?

Answer

Marrow clots within seconds without anticoagulant, so fresh smears must be prepared at once. Particles are spread thinly and air dried, giving good cell morphology.

#280 · Specimen Collection

What must be labelled on every bone marrow slide and tube at the bedside?

Answer

Two patient identifiers (name and ID number), the collection site, date and time, taken in front of the patient. Slides are labelled in pencil or solvent-resistant marker on the frosted end.

#281 · Specimen Collection

In what order are cerebrospinal fluid tubes numbered and used in most labs?

Answer

Tube 1 chemistry and serology, tube 2 microbiology, tube 3 cell count and differential (least blood contamination from the puncture), and tube 4 if sent for other tests.

#282 · Specimen Collection

Why is the CSF cell count tube usually the last or later tube collected?

Answer

The first tubes may carry blood from the needle trauma. A later tube is clearer, so the red and white cell counts better reflect the true fluid.

#283 · Specimen Collection

Why should a CSF specimen for cell count be delivered and counted promptly?

Answer

Cells, especially neutrophils, lyse within one to two hours, lowering counts. Deliver immediately at room temperature and never refrigerate a CSF intended for culture.

#284 · Specimen Collection

Which anticoagulant is acceptable for a synovial fluid cell count, and which should be avoided?

Answer

Use liquid EDTA or sodium heparin. Avoid powdered anticoagulants and oxalate, since they form crystals that mimic pathologic crystals under polarized light.

#285 · Specimen Collection

Why can a viscous synovial fluid cause trouble with a manual cell count, and how is it handled?

Answer

Hyaluronate makes the fluid thick, so cells clump. Pretreat with hyaluronidase or mix with a drop of saline as per procedure, then count in a hemocytometer.

#286 · Specimen Collection

When is a blood film for malaria best collected relative to a patient's fever?

Answer

Collect at the time of suspicion, ideally during or just after a fever spike when parasites are most numerous. Repeat films every 12 to 24 hours for three sets if the first is negative.

#287 · Specimen Collection

Why is a malaria film a STAT request, and how many films are needed to rule it out?

Answer

Plasmodium falciparum can be rapidly fatal, so it is read urgently. Negative thick and thin films on three sets over 24 to 48 hours are needed to reasonably exclude malaria.

#288 · Specimen Collection

Why should EDTA blood for a malaria film be smeared soon after collection?

Answer

Prolonged EDTA exposure distorts parasite and red cell morphology and can cause artifacts. Make thick and thin films within about an hour, ideally sooner.

#289 · Specimen Collection

How is plasma prepared for coagulation testing so that it is platelet-poor?

Answer

Centrifuge the citrate tube, commonly about 1500 g for 15 minutes, to give plasma with a platelet count below 10 x10^9/L. Do not disturb the buffy coat when removing plasma.

#290 · Specimen Collection

What is the purpose of a double spin when freezing plasma for later coagulation tests?

Answer

Spinning twice and transferring the plasma to a plastic tube removes residual platelets. Platelets would release phospholipid on freezing and thawing and shorten clotting times.

#291 · Specimen Collection

At what temperature should platelet-poor plasma be stored if frozen for later testing?

Answer

Freeze at minus 20 C for short storage (about two weeks) or minus 70 C for longer storage. Thaw quickly at 37 C and mix before testing, and never refreeze.

#292 · Specimen Collection

How are a STAT specimen and a routine specimen handled differently on arrival at the lab?

Answer

STAT samples are accessioned and tested immediately ahead of routine work, with results phoned if critical. Priority is set by clinical urgency, not by order of arrival.

#293 · Specimen Collection

What must be entered into the LIS when a specimen is accessioned?

Answer

Patient identifiers, ordering provider, tests requested, collection date and time, specimen type, collector ID and receipt time. A unique accession number links the tube to the record.

#294 · Specimen Collection

How does a patient's age affect interpretation of a hematology result, with one example?

Answer

Reference intervals differ by age. Newborns have higher hemoglobin and MCV than adults, and children have higher lymphocyte counts, so use age-specific reference ranges.

#295 · Specimen Collection

How do altitude and smoking change hemoglobin and hematocrit results?

Answer

Both raise them. Living at high altitude and smoking reduce oxygen delivery, so the body makes more red cells (smokers also have raised carboxyhemoglobin).

#296 · Specimen Collection

What must a collector do if a patient refuses a blood draw?

Answer

Respect the refusal, never force the draw, explain the reason for the test calmly, and notify the nurse or ordering provider. Document the refusal and who was informed.

#297 · Specimen Collection

What should be done when a tube is received with insufficient volume for the anticoagulant ratio, for example a short citrate tube?

Answer

Reject it and request recollection, since excess citrate dilutes the sample and falsely prolongs clotting times. Document the rejection and notify the collector or unit.

#298 · Specimen Collection

A tube label does not match the requisition. What is the correct action?

Answer

Do not test or relabel it. Return it for recollection, or follow the policy for irreplaceable samples with documented positive identification by the collector. Record the error and notify the supervisor.

#299 · Specimen Collection

Why must a light blue citrate tube be filled to the marked line?

Answer

Coagulation tests depend on a fixed 9 to 1 ratio of blood to 3.2 percent sodium citrate. An underfilled tube holds too much citrate, which over-dilutes the plasma and falsely prolongs PT and aPTT.

#300 · Specimen Collection

How should a capillary CBC sample be collected to limit platelet clumping?

Answer

Wipe away the first drop, let blood flow freely into the EDTA microtainer without squeezing, and mix by gentle inversion straight away. Slow collection or squeezing activates platelets and causes clumps and falsely low counts.

#301 · Reporting

Why does a hematology lab keep a written list of critical values with set limits?

Answer

Critical values signal possible immediate danger to the patient. The lab sets limits with its medical director and clinicians, so every technologist reports the same result the same way.

#302 · Reporting

A hemoglobin of 45 g/L is reported. What is the correct first action?

Answer

Treat it as a critical value: confirm the sample is correct and has no clot or dilution error, then phone the responsible clinician or nurse at once and document who was told, when, and the read-back.

#303 · Reporting

What must be documented when a critical value is phoned to a clinician?

Answer

The result, the date and time, the name of the person who received it, and that the value was read back. Many labs also record who made the call, to prove timely notification.

#304 · Reporting

A coagulation sample gives an aPTT that is greater than the instrument's upper limit. What do you check before reporting?

Answer

Check for a short draw (excess citrate), clotting, heparin contamination from a line draw, and a patient history of heparin. Then recollect if needed and phone the result if it is confirmed critical.

#305 · Reporting

Which two hematology results are classic critical values for bleeding risk when markedly low?

Answer

A very low platelet count and a very low fibrinogen. Both can mean spontaneous bleeding, so confirm the sample (check for clots, platelet clumping) and notify the clinician quickly.

#306 · Reporting

An analyzer gives a platelet count of 40 x10^9/L with a platelet clump flag. What should you do before reporting?

Answer

Examine the film for clumps and recollect in citrate if clumping is seen. Do not report the falsely low count as the true value. EDTA-dependent clumping is the usual cause.

#307 · Reporting

What should happen to a CBC that has an analyzer flag such as blasts or atypical lymphocytes?

Answer

Do not autoverify. Prepare and review a stained film, confirm or rule out the flagged cells, and add a verification comment or refer the film according to the lab's rules.

#308 · Reporting

What does a nucleated red blood cell flag do to the reported white cell count?

Answer

Nucleated RBCs can be counted as lymphocytes, falsely raising the WBC. The count must be corrected for the NRBCs, using the NRBC per 100 WBC count, before the corrected WBC is reported.

#309 · Reporting

The corrected WBC formula for nucleated RBCs: how is it applied?

Answer

Corrected WBC = observed WBC x 100 / (100 + NRBC per 100 WBC). Example: WBC 10.0 with 25 NRBC gives 10.0 x 100 / 125 = 8.0 x10^9/L.

#310 · Reporting

Why must hemoglobin and MCV reference intervals be age and sex specific?

Answer

Values change with development and hormones. Adult males run higher hemoglobin and hematocrit than adult females, and children and newborns have their own ranges. A single adult range would mislabel them.

#311 · Reporting

How do newborn hemoglobin and MCV compare with adult values?

Answer

Newborns have a high hemoglobin and a high (macrocytic) MCV, then both fall over the first weeks to months. An adult range applied to a newborn would wrongly flag normal babies.

#312 · Reporting

Which white cell type is normally most numerous in young children, in contrast to adults?

Answer

Lymphocytes. A child's differential is lymphocyte predominant until about school age, when neutrophils become the main cell, so adult differential ranges do not apply.

#313 · Reporting

What is a delta check in hematology?

Answer

A comparison of the current result with the patient's previous result, flagging a change larger than set limits. It can reveal a wrong patient, mislabelled sample, contamination, or a real clinical change.

#314 · Reporting

A delta check shows MCV changed from 88 fL to 112 fL in two days. What is the first thing to investigate?

Answer

Suspect a specimen identification error or mix-up first. Confirm the patient ID and label, then review other indices such as RDW and the film before reporting or phoning.

#315 · Reporting

A differential is reported as 60 percent neutrophils with a WBC of 5.0 x10^9/L. What is the absolute neutrophil count?

Answer

Absolute count = percentage x WBC. So 0.60 x 5.0 = 3.0 x10^9/L. Absolute counts are more meaningful than percentages when the total WBC is abnormal.

#316 · Reporting

Why can a normal differential percentage hide a real problem?

Answer

Percentages always total 100, so a normal looking percent can sit on a very low or high WBC. Converting to absolute counts shows true neutropenia, lymphopenia, or lymphocytosis.

#317 · Reporting

How is the absolute neutrophil count calculated when bands are reported?

Answer

ANC = WBC x (percent segmented neutrophils plus percent bands) / 100. Both mature and band forms are included, and the result is used to assess infection risk in neutropenia.

#318 · Reporting

What is the common grading scale for reporting abnormal red cell morphology?

Answer

A semi-quantitative scale from 1+ (slight) to 4+ (marked), based on how many cells per oil field show the change. Each lab defines its own thresholds and applies them consistently.

#319 · Reporting

Which standard terms describe a smear with variation in red cell size and in shape?

Answer

Anisocytosis describes variation in size, and poikilocytosis describes variation in shape. Use the standard terms, not informal descriptions, so every reader interprets the film the same way.

#320 · Reporting

Name three findings on a blood film that usually require referral to a pathologist or hematologist.

Answer

Blasts, schistocytes with thrombocytopenia (suspected TTP or HUS), and abnormal or unidentified cells such as suspected lymphoma cells or parasites. Follow the lab's written referral criteria.

#321 · Reporting

When blasts are seen on a smear from a patient with no prior history, what is the correct action?

Answer

Treat it as an urgent finding: confirm the cells, notify the pathologist or hematologist, and phone the clinician promptly as the lab policy requires. Do not wait for routine reporting.

#322 · Reporting

Schistocytes with a very low platelet count on a film suggest what urgent concern?

Answer

A microangiopathic process such as thrombotic thrombocytopenic purpura (TTP), which can be fatal without prompt treatment. Notify the clinician and pathologist immediately, in line with lab policy.

#323 · Reporting

Malaria parasites are seen on a thin film. How should the result be handled?

Answer

Treat it as an urgent critical result: phone the clinician right away, and report as the lab requires, since malaria is a reportable disease to public health. Do not delay for routine reporting.

#324 · Reporting

What is the usual therapeutic INR range for warfarin in most indications, such as atrial fibrillation and venous thromboembolism?

Answer

An INR of 2.0 to 3.0, with a target of 2.5. Some indications, such as certain mechanical heart valves, use a higher target set by the clinician, so report the result and let the clinician interpret.

#325 · Reporting

An INR comes back markedly above the lab's critical limit on a patient taking warfarin. What do you do?

Answer

Confirm the sample was filled correctly (a short draw falsely prolongs the INR) and has no clot, then phone the result to the clinician promptly and document the read-back, since bleeding risk is high.

#326 · Reporting

A PT and aPTT are both prolonged and do not correct on a 1:1 mixing study. How is this reported and what does it suggest?

Answer

Report the prolonged results with a comment that mixing did not correct. Failure to correct suggests an inhibitor (such as a lupus anticoagulant or factor antibody) rather than a factor deficiency.

#327 · Reporting

An aPTT is unexpectedly prolonged on a patient receiving unfractionated heparin. What comment or check supports accurate reporting?

Answer

Note the heparin therapy on the report. A prolonged aPTT is the expected effect. If the history is unclear, a thrombin time or heparin neutralization check can confirm heparin presence.

#328 · Reporting

Why can a lupus anticoagulant cause a prolonged aPTT in a patient who is not bleeding?

Answer

The antibody interferes with phospholipid-dependent clotting reagents in vitro. It is associated with thrombosis, not bleeding. Report the result with a comment recommending further investigation.

#329 · Reporting

A sample drawn from a central line is suspected of heparin contamination. What should the technologist do before reporting coagulation results?

Answer

Do not report the results as valid. Request a recollection by peripheral venipuncture, or from the line after proper discard, and document the reason for rejection.

#330 · Reporting

A coagulation tube is filled to only 50 percent of its draw volume. How is it handled?

Answer

Reject it and request a recollection. Underfilling alters the 9:1 blood to citrate ratio, excess citrate falsely prolongs PT and aPTT. Document the rejection.

#331 · Reporting

A visible clot is found in an EDTA tube for a CBC. What is the correct action?

Answer

Do not report the CBC. Clots trap cells and falsely lower counts. Request a recollection and record the rejection and the notification of the collector or ordering provider.

#332 · Reporting

A CBC specimen is markedly hemolyzed. Which results are affected and what comment is added?

Answer

In vitro hemolysis lowers the RBC count and hematocrit while hemoglobin stays accurate, so MCHC rises falsely. Add a specimen quality comment or request a recollection per laboratory policy.

#333 · Reporting

What is autoverification in hematology?

Answer

Computer rules compare instrument results with set limits, delta checks, flags and critical values. Results that pass all rules release without technologist review. Any failure holds the result for manual review.

#334 · Reporting

Name three typical triggers that stop a CBC from autoverifying.

Answer

Instrument suspect flags (such as blasts or abnormal WBC scattergram), failed delta check, and results outside critical or analytical limits. Any of these send the sample to smear review.

#335 · Reporting

Why must autoverification rules be validated before use and rechecked after changes?

Answer

Validation proves the rules hold back every unsafe result and release only reliable ones. Record the testing, and revalidate after any instrument, software or rule change, per ISO 15189 quality requirements.

#336 · Reporting

A hemoglobin result has been released and then found to be wrong. What is the first step?

Answer

Verify the correct result, then notify the ordering provider promptly if the error could affect care. Issue a corrected report that keeps the original visible and is clearly marked as corrected.

#337 · Reporting

What must a corrected laboratory report show?

Answer

The corrected result clearly labelled as corrected, the original result retained for reference, the date and time of correction, who made it, and the notification record. Never silently overwrite the original value.

#338 · Reporting

Who must be told about a corrected result that may have changed treatment?

Answer

The ordering provider or responsible clinician, notified directly. Record the name of the person told, the time, and who made the call, and follow the lab's incident or occurrence reporting process.

#339 · Reporting

What is a delta check and why is it useful on a cumulative report?

Answer

It compares a patient's current result with their previous one. A change beyond a set limit may signal mislabelling, contamination or a real clinical change, and it prompts review before release.

#340 · Reporting

A patient's hemoglobin falls from 130 g/L to 80 g/L over two days. How does a cumulative report help?

Answer

Trending shows a clear acute drop that a single result would hide. It supports clinical recognition of bleeding or hemolysis, and confirms the result is real rather than a labelling error.

#341 · Reporting

How is a white cell count corrected when nucleated red blood cells (nRBC) are present?

Answer

Corrected WBC = measured WBC x 100 divided by (nRBC per 100 WBC + 100). Example: 10.0 with 25 nRBC gives 10.0 x 100 / 125 = 8.0 x10^9/L.

#342 · Reporting

Why do nRBCs falsely raise the automated WBC count?

Answer

Nucleated red cells are counted as leukocytes by many analyzers because they have nuclei that survive lysis. Counting them per 100 WBC on the smear allows the corrected WBC to be reported.

#343 · Reporting

When is the nRBC correction applied to a manual or automated WBC?

Answer

When nRBCs are seen on the differential, per laboratory policy, commonly when they number 5 or more per 100 WBC. Report the corrected WBC and the nRBC count.

#344 · Reporting

Why are body fluid cell counts reported as nucleated cells with a differential, not as a blood CBC?

Answer

Body fluids hold few cells and different cell types. Report total nucleated cell count and RBC count per litre, plus the differential in percent, with the fluid type and specimen quality noted.

#345 · Reporting

A cerebrospinal fluid (CSF) cell count is ordered. What is the key handling and reporting point?

Answer

CSF is an urgent specimen and cells lyse quickly, so count promptly. Report cell counts and differential, and phone abnormal results such as organisms or blasts as critical results.

#346 · Reporting

Which unit and numerical pair is correct for a normal adult hemoglobin report in Canada?

Answer

Hemoglobin is reported in g/L (for example 140 g/L), not g/dL. A g/dL value of 14.0 entered as g/L is a classic 10-fold unit error that must be caught.

#347 · Reporting

What units are used for RBC count, hematocrit and MCV in Canadian hematology reports?

Answer

RBC in x10^12/L, hematocrit as a decimal fraction in L/L (for example 0.42), and MCV in fL. MCH is reported in pg and MCHC in g/L.

#348 · Reporting

How is a critical hematology result communicated verbally?

Answer

Phone the responsible clinician directly, give the patient identifiers and result, have the receiver read it back, then document who was told, the time, and the read-back. Escalate if there is no response.

#349 · Reporting

What is the absolute reticulocyte count formula and a typical use?

Answer

Absolute retic (x10^9/L) = retic percent x RBC count (x10^12/L) x 10. Example: 2.0 percent x 4.0 x 10 = 80 x10^9/L. It shows marrow response better than percent alone.

#350 · Reporting

Why report both retic percent and absolute count in an anemic patient?

Answer

Percent can look high only because RBCs are low. The absolute count shows true marrow output. A low absolute count with anemia suggests inadequate production.

#351 · Reporting

In what units is the Westergren erythrocyte sedimentation rate reported, and after what time is it read?

Answer

Report in mm/h, read at the 1 hour mark as the distance the red cell column falls below the plasma meniscus. Always record the unit with the number.

#352 · Reporting

What blood to anticoagulant ratio is used for a Westergren ESR collected in sodium citrate?

Answer

Four parts blood to one part 3.2 percent sodium citrate (4:1). Underfilling dilutes the sample incorrectly, so reject or recollect when the fill volume is wrong.

#353 · Reporting

Name two technical errors that falsely change an ESR result and the direction of the error.

Answer

A tube tilted from vertical falsely raises the ESR. Air bubbles, vibration or a warm bench also distort it. Refrigerated blood must return to room temperature before testing.

#354 · Reporting

An automated platelet count is unexpectedly low and the analyzer flags clumps. What is the first step before reporting?

Answer

Examine the smear for platelet clumps and fibrin. If clumps are present, do not report the low count as true. Recollect and repeat before releasing a result.

#355 · Reporting

Which tube is used to recollect a sample when EDTA-induced platelet clumping (pseudothrombocytopenia) is suspected?

Answer

A sodium citrate tube. The citrated count is then corrected for the 1:10 dilution (multiply by 1.1) and the report is annotated that it was a citrate count.

#356 · Reporting

How is a platelet count estimated from a blood smear, and when is it used?

Answer

Average the platelets per oil immersion field over several fields and multiply by the lab's validated factor (often 20 x10^9/L). Used to confirm or verify an instrument count.

#357 · Reporting

In newborn screening by HPLC, what does an FS pattern with no hemoglobin A suggest?

Answer

Possible sickle cell disease (or sickle beta zero thalassemia). It is only a screening result, so the report states confirmatory testing and clinical follow-up are required.

#358 · Reporting

What does an FAS pattern on newborn screening usually indicate, and how should the report be worded?

Answer

Usually sickle cell trait (carrier). Word it as consistent with trait, refer to the provincial program's follow-up pathway, and avoid calling it a disease.

#359 · Reporting

What is a basic rule for adding a test onto a sample already in the hematology lab?

Answer

Confirm the analyte is stable for the time elapsed since collection, the sample volume and storage are acceptable, then document who requested it and when.

#360 · Reporting

Why is an add-on manual differential or smear review often limited by sample age?

Answer

Blood in EDTA develops cell morphology artifacts over time (vacuolated neutrophils, crenated red cells). Past the lab's stability limit, a new sample is needed.

#361 · Reporting

What is reflex testing in hematology, and who approves the criteria?

Answer

A pre-approved rule that triggers a further test automatically when a result meets set criteria, such as a smear review for a flagged CBC. The medical director approves the rules.

#362 · Reporting

What should a technologist do on discovering a hematology result already released was wrong?

Answer

Notify the supervisor and the clinician right away, issue a corrected report clearly marked as corrected, keep the original visible in the record, and file an occurrence report.

#363 · Reporting

What is a near miss in laboratory reporting, and why is it still reported?

Answer

An error caught before it reached the patient or clinician. It is reported so the system can find trends and fix the cause before a real harm event occurs.

#364 · Reporting

When a reporting error reached the patient chart, what does disclosure involve?

Answer

The responsible clinician and lab leadership inform the patient or care team honestly about what happened and what was corrected, following the facility's disclosure policy.

#365 · Reporting

What must an audit trail for a hematology result show when a result is edited?

Answer

Who made the change, the date and time, the original value, the new value, and the reason. Original results must never be deleted or overwritten without a trace.

#366 · Reporting

How long must hematology records be kept?

Answer

It depends on provincial regulation and the lab's accreditation policy, so follow your lab's written retention schedule, which sets different periods for reports, QC records and slides.

#367 · Reporting

Under Ontario's PHIPA, who may receive a patient's hematology result?

Answer

The ordering provider and others in the patient's circle of care, or the patient through the lab's release process. Other provinces have their own health privacy laws.

#368 · Reporting

A caller asks for a patient's CBC result by phone. What should you do before releasing it?

Answer

Verify who the caller is and that they are authorized, using the lab's procedure and at least two patient identifiers. If unsure, refer them to the ordering provider.

#369 · Reporting

Why should hematology results not be shared with a patient's family member without consent?

Answer

Health information is protected by provincial privacy law. Release requires the patient's consent or another lawful authority, and the lab's policy decides how this is checked.

#370 · Reporting

What may an MLT add as an interpretive comment on a hematology report?

Answer

Standardized descriptive comments approved by the medical director, such as 'platelet clumps seen, recollection suggested'. The comment describes findings and does not diagnose.

#371 · Reporting

What is outside an MLT's scope when reporting abnormal hematology findings?

Answer

Making a diagnosis or recommending treatment. The MLT reports findings, flags them using lab policy, and refers interpretation to the pathologist or physician.

#372 · Reporting

Which assay is commonly used to monitor unfractionated heparin by anti-Xa, and what is a typical therapeutic range?

Answer

A chromogenic anti-Xa assay on citrated platelet poor plasma. A typical UFH therapeutic range is 0.3 to 0.7 IU/mL, but use the range set by your lab.

#373 · Reporting

What can falsely affect an anti-Xa result, and how should it be handled before reporting?

Answer

Marked hemolysis, icterus or lipemia can interfere with the chromogenic reading. Check the sample, note the interference or request recollection per lab policy.

#374 · Reporting

You find blasts on a smear from an unexpected CBC. How should this be communicated?

Answer

Telephone the responsible clinician promptly, use read back, document the name and time of the call, and refer the smear to the pathologist per the lab's critical result policy.

#375 · Reporting

What must a technologist document when a critical hemoglobin is phoned to a clinician?

Answer

The result, the date and time, the full name of the person who received it, and that the value was read back. This record proves the critical result reached someone able to act on it.

#376 · Quality Assurance

On a Levey-Jennings chart, which lines are normally drawn, and what do they represent?

Answer

The mean as the centre line, plus lines at plus and minus 1, 2 and 3 SD. Each control result is plotted in run order so shifts, trends and outliers are visible at a glance.

#377 · Quality Assurance

How is the mean and SD for a Levey-Jennings chart usually established for a new control lot in your own lab?

Answer

Run the new control repeatedly (commonly 20 or more data points over several days), calculate the lab's own mean and SD, then compare with the manufacturer's assay range before using it for QC.

#378 · Quality Assurance

A control result falls 2.5 SD above the mean on a Levey-Jennings chart. Is it automatically out of control?

Answer

No. Beyond 2 SD is a warning only. About 5 percent of values fall outside plus or minus 2 SD by chance, so it triggers a check of other Westgard rules, not rejection alone.

#379 · Quality Assurance

What does the Westgard 1-3s rule detect, and is it a warning or rejection rule?

Answer

One control result beyond plus or minus 3 SD. It is a rejection rule that mainly flags random error (or a large systematic error). Stop, troubleshoot and do not report patient results.

#380 · Quality Assurance

What does the 2-2s rule mean, and what kind of error does it usually indicate?

Answer

Two consecutive control results exceed the same limit of plus 2 SD or minus 2 SD. It is a rejection rule that mainly indicates systematic error such as calibration drift or reagent deterioration.

#381 · Quality Assurance

How is the R-4s rule applied to two levels of control, and which error does it detect?

Answer

One control is above plus 2 SD and the other is below minus 2 SD within the same run, a range of more than 4 SD. It detects random error (imprecision).

#382 · Quality Assurance

What does the 4-1s rule signal on a CBC or coagulation control chart?

Answer

Four consecutive control results exceed 1 SD on the same side of the mean. It signals systematic error, often an early shift or drift, before results become clearly unacceptable.

#383 · Quality Assurance

What does the 10-x rule indicate, and how does it differ from a trend?

Answer

Ten consecutive control values fall on the same side of the mean: a systematic shift. A trend instead is a gradual steady rise or fall over about six or more points.

#384 · Quality Assurance

A hematology control chart shows an abrupt move of all points to a new level. Is this a shift or a trend, and what are common causes?

Answer

A shift. Common causes: new reagent or control lot, recalibration, new lamp or aperture change, or maintenance. A trend by contrast suggests gradual deterioration such as reagent or control aging.

#385 · Quality Assurance

Which type of error is revealed by a wide scatter of control results around an unchanged mean?

Answer

Random error (imprecision), shown by increased SD. Typical causes are bubbles, clogged aperture, inconsistent pipetting or mixing, or unstable reagent. It is not corrected by recalibration.

#386 · Quality Assurance

Define accuracy versus precision in laboratory testing.

Answer

Accuracy is closeness of a result to the true value (low bias). Precision is the reproducibility of repeated results (low SD or CV). A method can be precise yet inaccurate.

#387 · Quality Assurance

How is the coefficient of variation (CV) calculated? Give a short worked step.

Answer

CV percent = (SD divided by mean) x 100. Example: mean WBC 8.0 x10^9/L, SD 0.2 gives (0.2 / 8.0) x 100 = 2.5 percent. It allows comparing precision across different levels.

#388 · Quality Assurance

How is percent bias calculated when comparing a lab's mean to a target value?

Answer

Bias percent = ((lab mean minus target) divided by target) x 100. Example: lab mean 4.2, target 4.0 gives plus 5 percent. Bias measures systematic error and relates to accuracy.

#389 · Quality Assurance

Why are three levels of CBC control (low, normal, high) used rather than one?

Answer

They verify performance across the reportable range, including clinically critical low and high values, and help detect problems that affect only part of the range, such as linearity or carryover.

#390 · Quality Assurance

What handling steps matter for commercial CBC control blood before running it?

Answer

Warm to room temperature, mix gently by rolling and inversion per the insert (not vigorous shaking), run promptly, and return to refrigerator. Respect open-vial stability. Poor mixing causes false imprecision.

#391 · Quality Assurance

Before a new lot of CBC controls replaces the current lot, what is the best practice?

Answer

Run the new lot in parallel with the current lot, collect enough data points to set the lab's own mean and SD, then confirm values fall within the manufacturer's range before the changeover.

#392 · Quality Assurance

What is patient-based QC using the moving average of red cell indices (Bull's algorithm)?

Answer

Averaging MCV, MCH and MCHC over batches of consecutive patient results (traditionally 20). Drift in the running average, plotted like a control chart, signals a calibration or analyzer problem independent of commercial controls.

#393 · Quality Assurance

What is a delta check and what does a failed delta check on a CBC prompt?

Answer

A comparison of the current result with the same patient's previous result for an unexpected change. A failure prompts checking for specimen mislabelling, wrong-blood-in-tube, clots or true clinical change before reporting.

#394 · Quality Assurance

Which red cell index is considered most stable and useful in moving-average and delta checks, and why?

Answer

MCV, because it is measured directly and varies little in an individual. A sudden MCV change between samples from one patient strongly suggests a specimen identification error.

#395 · Quality Assurance

What is the purpose of external proficiency testing (PT) in hematology, and what formats are commonly included?

Answer

It compares your results with peer labs and reference values using unknown samples to assess accuracy. Hematology surveys typically include CBC, differential, blood film images and coagulation tests.

#396 · Quality Assurance

A lab fails a proficiency testing challenge. What is the proper response?

Answer

Investigate the root cause, document corrective action, recheck or retest as appropriate, assess any effect on patient results, and keep records. Never send PT samples to another lab for confirmation.

#397 · Quality Assurance

What is the difference between method validation and method verification?

Answer

Validation proves a new or modified method meets its intended use (full study). Verification confirms an established, manufacturer-cleared method performs as claimed in your own lab, typically precision, accuracy and reportable range.

#398 · Quality Assurance

How is carryover on a hematology analyzer commonly evaluated?

Answer

Run a high-concentration sample three times, then a low-concentration sample three times. Carryover percent = (low1 minus low3) divided by (high3 minus low3) x 100, compared with the manufacturer's limit.

#399 · Quality Assurance

What is calibration verification and when is it required for a hematology analyzer?

Answer

Running materials of known value to confirm the existing calibration still holds across the reportable range. It is done at set intervals and after major maintenance, component replacement, or when QC shows persistent bias.

#400 · Quality Assurance

When a new thromboplastin reagent lot is introduced, what must be re-established or verified, and why?

Answer

Verify the ISI (from the manufacturer) and establish the lab's mean normal PT using fresh normal plasmas. INR = (patient PT / mean normal PT) raised to the ISI, so errors in either value shift INR results.

#401 · Quality Assurance

What is the usual minimum number of reference individuals recommended by CLSI EP28-A3c to establish a new reference interval directly?

Answer

At least 120 reference individuals per partition. The 2.5th and 97.5th percentiles are then taken as the limits, so the central 95% of healthy values defines the interval.

#402 · Quality Assurance

How many reference specimens does a laboratory typically test to verify a manufacturer's reference interval, and what passes?

Answer

Usually 20 healthy individuals. Verification passes if no more than 2 of the 20 results (10%) fall outside the stated limits. More than 2 outside means investigate or establish your own.

#403 · Quality Assurance

Why are percentiles, not mean plus or minus 2 SD, often used to set a hemoglobin or WBC reference interval?

Answer

Many hematology analytes are not normally distributed (WBC is skewed). A nonparametric 2.5th to 97.5th percentile does not assume a Gaussian shape, so it avoids a distorted lower or upper limit.

#404 · Quality Assurance

Why might a hemoglobin reference interval need separate partitions?

Answer

Hemoglobin differs by sex, age, and for some populations pregnancy or altitude. Healthy adult males and females, children and newborns each need their own interval, or abnormal results will be misflagged.

#405 · Quality Assurance

What is the purpose of a method comparison study when a new hematology analyzer replaces the old one?

Answer

To show the new instrument gives clinically equivalent results to the current one across the measuring range. It uses fresh patient samples run on both, then regression or bias plots to judge agreement.

#406 · Quality Assurance

In a manual versus automated differential comparison, what does a correlation coefficient alone fail to show?

Answer

Correlation shows how closely results move together, not whether they agree. A constant or proportional bias can exist with r near 1.0, so slope, intercept and mean bias must also be examined.

#407 · Quality Assurance

Why is the manual 100-cell differential less precise for rare cell types than an automated 10 000-cell count?

Answer

Counting only 100 cells gives wide statistical error, especially for low-frequency cells like basophils or blasts. Automated counting samples far more cells, so the percentage estimate is much tighter.

#408 · Quality Assurance

How often should Wright-Giemsa stain quality be checked, and what is assessed?

Answer

Check with each new stain batch and whenever staining looks off, and review daily slides. Look for pink RBCs, blue-purple nuclei with clear chromatin, correct granule colour, and no precipitate or background stain.

#409 · Quality Assurance

A blood smear shows excessively blue RBCs and a blue background. What stain problem is the likely cause?

Answer

The stain or buffer pH is too alkaline, or the slide was stained or left too long. Check buffer pH (about 6.8 to 7.2), staining time and rinse. Restain or prepare a fresh smear.

#410 · Quality Assurance

What does morphology slide review competency typically involve for a technologist?

Answer

Reviewing a set of known or previously reviewed slides and correctly identifying cells and abnormalities, compared with an expert's answers. Results are scored and gaps trigger retraining before independent reporting.

#411 · Quality Assurance

What is the first action when hematology QC results fall outside acceptable limits?

Answer

Stop reporting patient results for that analyte. Do not release anything until the cause is found and QC is acceptable, then review patient results run since the last acceptable QC.

#412 · Quality Assurance

Name a logical first troubleshooting step after a QC failure on an automated analyzer.

Answer

Check for obvious causes first: expired or mishandled control, reagent level, recent maintenance or calibration, then repeat with a fresh vial of control. Change one variable at a time.

#413 · Quality Assurance

After a QC failure is corrected, which patient samples may need to be repeated?

Answer

Patient samples tested since the last acceptable QC, judged by the lab's SOP and the shift in results. Samples near decision limits or showing a clinically significant difference are reanalyzed, and corrected reports issued.

#414 · Quality Assurance

Give one preanalytical error that falsely lowers a platelet count.

Answer

Platelet clumping from a clotted or EDTA-dependent sample, or a short draw. A small clot traps platelets. Check for clots, review the smear for clumps, and consider a citrate tube recollection.

#415 · Quality Assurance

Which type of error is incorrect specimen labelling, and why does it matter most?

Answer

A preanalytical error, since it causes wrong-patient results. A mislabelled specimen can pass all analytical QC, so patient identification checks at collection and receipt are the main defense.

#416 · Quality Assurance

Give an example of an analytical error source in hematology.

Answer

Instrument problems such as a blocked aperture, reagent deterioration or calibration drift. These show as QC shifts or trends, flagged by Westgard rules, and are caught before patient results are released.

#417 · Quality Assurance

Give an example of a postanalytical error in hematology reporting.

Answer

Transcription error, wrong units, missed critical value notification or results sent to the wrong chart. Verify manual entries, use critical value policies, and confirm LIS interfaces after changes.

#418 · Quality Assurance

What is ISO 15189 and who uses it?

Answer

An international standard for quality and competence of medical laboratories. Canadian labs use it, with provincial programs often built on it, to show technical competence and a working quality management system.

#419 · Quality Assurance

What does accreditation by a provincial body typically require of a hematology lab?

Answer

Documented quality management, validated methods, QC and proficiency testing, competent trained staff and regular assessment visits. Requirements vary by province, so follow the specific program the lab belongs to.

#420 · Quality Assurance

What is the aim of occurrence (incident) management in a laboratory?

Answer

To record errors and near misses, correct them and prevent repeats, not to assign blame. Reports are investigated, trended and closed with documented corrective and preventive action.

#421 · Quality Assurance

What is the idea behind root cause analysis after a repeated specimen mix-up?

Answer

Ask why until the system cause is found, such as a confusing workflow or lack of a second identifier check, rather than stopping at 'technologist error'. Fix the process so it cannot happen again.

#422 · Quality Assurance

Why must SOPs in a hematology lab go through document control?

Answer

So only the current, approved version is in use. Each SOP has an owner, version, effective date and authorized review, and obsolete copies are removed from the bench.

#423 · Quality Assurance

What does change control require before a new reagent lot or procedure change goes live?

Answer

The change is assessed, approved, verified with testing as needed (for example lot-to-lot checks), documented, and staff are trained before use. Records show who approved it and when.

#424 · Quality Assurance

How is hematology technologist competency usually assessed and documented, including for new hires?

Answer

Direct observation, review of results, unknown samples and problem solving, at orientation, again at six months, then at least annually. Records are kept with training files and gaps trigger retraining.

#425 · Quality Assurance

What QC is needed for a point-of-care INR meter?

Answer

Run liquid or built-in electronic controls as the manufacturer and lab policy require, and compare with a lab INR periodically. Operators must be trained, and results outside limits stop patient testing.

#426 · Equipment

Which lens cleaner and paper should be used to remove immersion oil from a microscope oil objective at the end of the day?

Answer

Wipe the 100x objective with clean lens paper, then lens paper lightly moistened with lens cleaner. Never use tissue or gauze, which scratch the lens, and never leave oil to dry on it.

#427 · Equipment

What is the purpose of Kohler illumination, and what does the technologist adjust to set it up?

Answer

It gives even, glare-free field illumination with maximum resolution. Focus the specimen, close and centre the field diaphragm image, then adjust the condenser height and aperture diaphragm.

#428 · Equipment

A microscope image is dim and uneven after the lamp was replaced. What is the first maintenance check?

Answer

Check that the new bulb is seated and centred (aligned) and that the condenser and diaphragms are set correctly. Record the lamp change and hours of use in the maintenance log.

#429 · Equipment

What does a hematology analyzer background (blank) count check confirm each day, and what result is expected?

Answer

It confirms the diluent and fluidics are clean and free of contamination or carryover. Counts should be at or below the manufacturer's stated limits, such as WBC near zero, before running patients.

#430 · Equipment

Why are daily startup quality control runs done on a hematology analyzer before patient samples are reported?

Answer

They verify the instrument is calibrated and performing within acceptable limits after startup. Run at least normal, low and high controls as per lab policy, and release results only if QC passes.

#431 · Equipment

What is the usual purpose of the daily or scheduled automated cleaning cycle (such as a bleach or enzymatic cleaner) on a hematology analyzer?

Answer

It removes protein build up and residue from tubing, flow cells and apertures that can cause clogs, high background counts and erroneous results. Follow the manufacturer's schedule and document it.

#432 · Equipment

What must be done before a centrifuge is started, to prevent rotor damage and unsafe vibration?

Answer

Balance the tubes opposite each other by equal volume and mass, use matching tube types, and close and latch the lid. An unbalanced load can cause serious mechanical failure and aerosol release.

#433 · Equipment

How is the speed of a centrifuge verified, and how often should this be checked?

Answer

Use a tachometer (or a calibrated strobe) to compare actual rpm against the dial setting, and a calibrated timer to check run time. Check at least as often as the lab quality program requires, commonly every 6 to 12 months.

#434 · Equipment

Why is the microhematocrit centrifuge timing and speed verification important, and what happens if the spin time is too short?

Answer

Insufficient spin leaves red cells incompletely packed, falsely raising the hematocrit reading from trapped plasma. Verify the time needed to reach maximum packing and recheck periodically.

#435 · Equipment

What common daily maintenance applies to a coagulation analyzer's reagent handling to avoid erroneous PT and aPTT results?

Answer

Reconstitute and store reagents as directed, check expiry and on board stability times, bring to the required temperature, and never mix lots without validation. Label open dates on all vials.

#436 · Equipment

What routine checks keep a coagulation analyzer's sample probe working correctly?

Answer

Inspect the probe for clots, bends and build up, clean it with the recommended cleaning solution, and confirm it is aligned to pierce the tube and cuvette accurately. Problems cause short sampling and false results.

#437 · Equipment

Why must the incubation temperature of a coagulation analyzer be monitored, and what temperature is typical for clotting assays?

Answer

Clotting enzyme reactions are temperature dependent, so shifts change clotting times. Most coagulation assays are run at 37 degrees C, with the temperature checked and recorded per the lab's schedule.

#438 · Equipment

How is a pipette verified for accuracy, and which tool is commonly used?

Answer

Dispense a known volume of water onto a calibrated analytical balance, weigh it repeatedly, and compare to the expected mass (1 mL of water weighs about 1 g). Calculate accuracy and precision against tolerance.

#439 · Equipment

What daily or routine verification is required for an analytical balance used in the laboratory?

Answer

Check it is level, clean and zeroed, then weigh certified check weights and record results. Service and external calibration are done at set intervals by an approved provider.

#440 · Equipment

What temperature range should a laboratory reagent refrigerator be kept in, and how is it monitored?

Answer

Keep it between 2 and 8 degrees C. Use a calibrated thermometer, record the temperature daily (or use continuous logging), and document corrective action for any out of range reading.

#441 · Equipment

A heat block used for incubation reads 3 degrees C below its set point. What is the correct action?

Answer

Do not use it for the test. Verify with a calibrated reference thermometer, adjust or take it out of service, and document the problem and corrective action. Repeat any affected tests.

#442 · Equipment

What is the correct way to clean a hemocytometer and its coverslip after use?

Answer

Rinse the chamber gently with water, dry with lens paper and never scrub the ruled area. Use a thick, flat, clean coverslip made for the chamber, as scratches and residue distort counts.

#443 · Equipment

How can you tell a hemocytometer coverslip is correctly seated before counting cells?

Answer

Look for Newton's rings (rainbow interference lines) where the coverslip contacts the chamber rails. Their presence shows the proper chamber depth of 0.1 mm has been achieved.

#444 · Equipment

What routine maintenance prevents poor smears on an automated slide maker or stainer?

Answer

Replace stains and buffers on schedule, filter or change reagents as per the manufacturer, and clean the spreader and staining baths. Check smear and stain quality with a control slide before patients.

#445 · Equipment

What is a cytocentrifuge used for in a hematology or body fluid lab, and what is a key setup point?

Answer

It deposits cells from low cellularity fluids (such as CSF) onto a slide as a monolayer. Use the correct filter card, slide and volume, and balance the cytospin heads before running.

#446 · Equipment

Which daily checks does a flow cytometer need before running patient samples?

Answer

Run quality control beads to verify laser alignment, fluidics and detector performance, and check sheath fluid and waste levels. Results must meet acceptance limits before patient data are reported.

#447 · Equipment

What are calibration beads used for on a flow cytometer?

Answer

They set and verify instrument settings, such as voltages and fluorescence sensitivity, and confirm compensation and consistency over time. Beads of known fluorescence give a reproducible reference signal.

#448 · Equipment

What is the first thing a technologist should do when an analyzer fails and cannot be used?

Answer

Remove it from service with a clear label, notify the supervisor, and switch to the backup instrument or procedure. Record the failure, time and actions in the equipment log.

#449 · Equipment

After repairing an instrument that was down, what must happen before patient testing resumes?

Answer

Verify performance: run QC and, as the lab policy requires, recalibrate and compare patient samples with a backup method. Document the repair, the verification results and the supervisor's approval.

#450 · Equipment

How often should the microhematocrit centrifuge speed and timer be verified?

Answer

At the interval written in the laboratory procedure, using a tachometer and a calibrated timer, with the result recorded. Too little speed or time leaves red cells packed incompletely and gives a falsely high hematocrit.

#451 · Safety

Which two WHMIS 2015 pictograms would you expect on a methanol-based Wright-Giemsa stain bottle?

Answer

Flame (flammable liquid) and skull and crossbones (acute toxicity), often with the health hazard pictogram too. Methanol is flammable and toxic by ingestion, skin contact and inhalation, so the label must show both.

#452 · Safety

Under WHMIS 2015, how many sections does a safety data sheet (SDS) have, and which section covers first aid?

Answer

An SDS has 16 sections in a standard order. Section 4 is first aid measures. Section 8 covers exposure controls and PPE, and Section 7 covers handling and storage.

#453 · Safety

A stain is poured from a stock bottle into a smaller working bottle. What must the new container carry under WHMIS 2015?

Answer

A workplace label with the product identifier, safe handling information, and a reference to the SDS. Decanted containers cannot go unlabelled, even for short use, unless the contents are used immediately by the person who filled it.

#454 · Safety

What is the minimum PPE for routine handling of whole blood specimens in the hematology bench area?

Answer

Fluid-resistant lab coat or gown, gloves, and eye and face protection when splashes are possible. Routine practices treat all blood as potentially infectious, so protection is applied to every specimen, not only known positives.

#455 · Safety

In what order is PPE removed after bench work to limit contamination?

Answer

Remove gloves first (they are the most contaminated), then eye protection, then the lab coat, and finally perform hand hygiene. Peel gloves inside out and never touch bare skin with a contaminated glove.

#456 · Safety

When must gloves be changed during hematology work, and what must never be worn out of the lab?

Answer

Change gloves when torn, visibly contaminated, or when moving between tasks. Never wear gloves into hallways, offices, or break rooms, and never touch telephones, door handles or keyboards with soiled gloves.

#457 · Safety

Why should methanol-based fixatives and stains be stored away from oxidizers such as bleach and nitric acid?

Answer

Flammable organics can react violently or ignite when in contact with oxidizers. Segregate by hazard class: flammables in a flammable cabinet, oxidizers separate, acids and bases apart from each other.

#458 · Safety

Where should acids and bases be stored in relation to each other and the floor?

Answer

Store acids and bases in separate compartments or secondary containment, with large containers low on a shelf, below eye level. Mixing them through a leak can release heat and hazardous fumes.

#459 · Safety

Why keep only a small working volume of flammable stain on the bench and store the rest in a flammable cabinet?

Answer

Keep only the small working amount on the bench and store the rest in a labelled, approved flammable cabinet. This limits fuel available to a fire and protects containers from heat and ignition sources.

#460 · Safety

What is the first step when a small volume of methanol-based stain spills on the bench?

Answer

Alert nearby staff, remove ignition sources, check the SDS, and put on suitable PPE. Then contain the spill with absorbent material designed for solvents and dispose of it as chemical waste.

#461 · Safety

How long should you flush eyes at an eyewash station after a chemical splash?

Answer

Flush for at least 15 minutes, holding the eyelids open and rolling the eyes. Start immediately, and seek medical attention afterward, taking the SDS along.

#462 · Safety

A staff member's clothing is soaked with a corrosive chemical. What is the correct action?

Answer

Go straight to the safety shower, remove contaminated clothing while under the water, and flush the skin for at least 15 minutes. Call for help and report the incident through the occupational health process.

#463 · Safety

What is the immediate response to a needlestick injury with a contaminated lancet?

Answer

Wash the site with soap and running water, do not squeeze or scrub, and cover it. Report to the supervisor right away and attend occupational health or emergency for risk assessment and post-exposure prophylaxis if indicated.

#464 · Safety

What information about the source patient is needed after a blood exposure incident?

Answer

The source's bloodborne virus status (HIV, hepatitis B, hepatitis C), obtained through the proper consent and testing process by the responsible clinician. This guides prophylaxis decisions, and timing matters, particularly for HIV.

#465 · Safety

What is the principle of routine practices in Canadian health care?

Answer

Treat the blood and body fluids of every patient as potentially infectious, and apply the same barrier precautions and hand hygiene to all. It replaces the older approach of relying on known infection status.

#466 · Safety

Why must needles never be recapped by hand, bent or broken before disposal?

Answer

Recapping and manipulating needles is a leading cause of needlestick injuries. Discard sharps immediately after use, uncapped, into a nearby puncture-resistant container, using a safety device when available.

#467 · Safety

At what fill level should a sharps container be closed and replaced?

Answer

When it is about three quarters full (at the fill line). Overfilling forces fingers near the opening and raises needlestick risk. Never push sharps down or reach into the container.

#468 · Safety

What does the acronym RACE stand for in a lab fire emergency?

Answer

Rescue anyone in danger, Alarm (activate the pull station and call), Confine the fire by closing doors, then Extinguish if small and safe, or Evacuate. Your own safety comes before fighting a fire.

#469 · Safety

Which extinguisher class and the PASS steps apply to a small flammable liquid fire from a solvent?

Answer

Class B (flammable liquids), typically a dry chemical or CO2 unit. PASS means Pull the pin, Aim at the base, Squeeze the handle, Sweep side to side. Never use water on a solvent fire.

#470 · Safety

Which hematology procedure should be done in a Class II biological safety cabinet rather than on an open bench?

Answer

Procedures likely to produce aerosols or splashes with potentially infectious material, such as vortexing, opening pooled or suspect specimens, or preparing material from biohazardous cultures. The cabinet protects the worker, the specimen and the room.

#471 · Safety

Why is a biological safety cabinet not suitable for working with volatile toxic or flammable chemicals?

Answer

A recirculating cabinet can concentrate vapours inside and an ignition source may create a fire risk. Use a chemical fume hood, which vents vapours to the outside, for volatile solvents and fixatives.

#472 · Safety

How must a blood specimen be packaged when sent to another laboratory by road in Canada?

Answer

As a UN3373 Biological Substance, Category B under TDG: leakproof primary container, absorbent material, a sealed secondary container, and a rigid outer package with the correct marking. Staff must hold current TDG training.

#473 · Safety

What should be done if a specimen leaks in a transport bag when it arrives at the lab?

Answer

Wear gloves and eye protection, handle it in a contained way, and decontaminate the bag and surfaces with an approved disinfectant. Notify the sender and follow the lab's incident reporting and specimen rejection process.

#474 · Safety

What ergonomic adjustments reduce neck and back strain during prolonged microscope differentials?

Answer

Raise the microscope or adjust the eyepiece so the neck stays neutral, use an adjustable chair with back support, keep the elbows supported, and take short micro breaks. Avoid hunching toward the eyepieces.

#475 · Safety

What should an MLT do on noticing recurring wrist or hand pain from repetitive pipetting and slide work?

Answer

Report it early to the supervisor and occupational health, and complete an incident or injury report. Early reporting allows workstation changes and prevents a repetitive strain injury becoming chronic.

#476 · Professionalism

A colleague from another department asks what result a named patient had, out of curiosity. What should you do?

Answer

Decline politely. Patient information is shared only with those directly involved in that patient's care (circle of care). Curiosity is not a valid need to know, and looking it up is a privacy breach.

#477 · Professionalism

Which type of law governs how Canadian laboratories handle personal health information, and what does it generally require?

Answer

Provincial or territorial health privacy law (for example PHIPA in Ontario). It requires collecting, using and disclosing personal health information only as needed for care, with safeguards, and breaches must be reported.

#478 · Professionalism

How should you handle a specimen label or requisition with patient identifiers when disposing of paper waste?

Answer

Shred or use a secure confidential disposal bin. Never place identifiable paperwork in regular trash, because identifiers on paper are personal health information and carry the same privacy duty as electronic records.

#479 · Professionalism

Why must you log out of the laboratory information system when you leave a workstation?

Answer

Your login is tied to your identity. Anyone using an open session acts under your name, creating unauthorized access and an inaccurate audit trail. Lock or log out every time you step away.

#480 · Professionalism

What does scope of practice mean for a medical laboratory technologist?

Answer

The activities you are educated, authorized and competent to perform under your regulator's standards and your employer's policies. Tasks outside that scope must be referred to a qualified person, even if you are asked to do them.

#481 · Professionalism

You are unsure how to interpret an unusual result on a critical test. What is the appropriate first step?

Answer

Do not guess or release it. Review the procedure and QC, then escalate to a senior technologist, supervisor or pathologist per policy. Referring early protects the patient and keeps you within your competence.

#482 · Professionalism

A supervisor asks you to perform a procedure you have not been trained or assessed on. How should you respond?

Answer

Say that you are not yet competent in it and request training and a competency assessment first. Performing untrained procedures risks patient harm and falls outside what you can be accountable for.

#483 · Professionalism

You discover you released a wrong result to the chart. What does professional accountability require?

Answer

Report it right away to your supervisor, follow the correction and notification procedure so the clinician is told, and document the error honestly. Concealing or delaying disclosure increases harm and violates ethical standards.

#484 · Professionalism

What is a conflict of interest in laboratory practice? Give one example.

Answer

A situation where a personal interest could influence, or appear to influence, professional judgement. Example: accepting a gift from a reagent vendor while involved in choosing the next analyzer. Declare it to your supervisor.

#485 · Professionalism

Why is falsifying or back-filling a QC or maintenance record never acceptable?

Answer

Records must reflect what actually happened. Falsification is dishonest, hides real problems and can lead to unreliable patient results. It can also result in regulator discipline and loss of registration.

#486 · Professionalism

How do you confirm you have the correct patient before explaining a procedure to them?

Answer

Use at least two identifiers, such as full name and date of birth, asking the patient to state them and comparing with the requisition and wristband. Never rely on room number or ask leading yes or no questions.

#487 · Professionalism

A patient is anxious about a blood draw. What communication approach helps most?

Answer

Introduce yourself, explain the steps in plain language, listen to concerns and offer reassurance without promising no pain. Clear and calm communication builds trust and often reduces fear and movement during collection.

#488 · Professionalism

A family member asks you to explain a patient's results. How should you respond?

Answer

Do not interpret or release results. Explain that results go through the patient's care provider, and refer the family to the nurse or physician. Release to others requires the patient's consent or legal authority.

#489 · Professionalism

What are the key elements of an effective verbal critical value report to a nurse or physician?

Answer

Identify yourself, give the patient's two identifiers, the test, value and units, and the time. Request read-back to confirm, then document who received it and when, per lab policy.

#490 · Professionalism

What does a regulatory college generally expect of its registered technologists regarding continuing competence?

Answer

Ongoing participation in a quality assurance or professional development program, such as reflective learning plans, education hours or assessments, with records kept. Requirements vary by province, so follow your own college's program.

#491 · Professionalism

Why should a technologist keep a personal record of continuing education activities?

Answer

Regulators may audit or request proof of learning during registration renewal. Dated records of courses, in-services and reflections show you maintained competence and help you identify future learning needs.

#492 · Professionalism

What is the purpose of a structured handover between shifts in the laboratory?

Answer

To pass on pending work, instrument problems, critical results and unresolved issues clearly and completely. Written or verbal handover prevents lost information, repeated work and delayed patient results across shift changes.

#493 · Professionalism

Several urgent specimens arrive at once while routine work is waiting. How do you prioritize?

Answer

Process by clinical urgency per lab policy: STAT and critical specimens, such as emergency or time-sensitive samples, come first. Communicate expected delays to the requesting unit and ask for help if overloaded.

#494 · Professionalism

What does cultural safety mean in healthcare, and how does it apply to a laboratory visit?

Answer

Care in which the patient defines whether they feel respected and safe, free from bias and discrimination. Use respectful language, ask preferences, and recognize power imbalances, including the impact of historical harms on Indigenous peoples.

#495 · Professionalism

A patient who is hard of hearing needs a venipuncture. What accommodation is appropriate?

Answer

Face the patient, speak clearly at a normal pace, use written notes or gestures, and confirm understanding. Use a qualified interpreter where available. Do not assume understanding from a nod or shout.

#496 · Professionalism

A patient refuses to have blood drawn. What is the correct response?

Answer

Respect the refusal, because competent patients have the right to decline. Explain the purpose and consequences calmly, do not coerce or restrain, and notify the nurse or physician and document the refusal.

#497 · Professionalism

What makes consent valid for a specimen collection procedure?

Answer

It must be voluntary, informed and given by a capable person. Implied consent applies to routine procedures, such as an outstretched arm, but must be reconfirmed if the patient hesitates or objects.

#498 · Professionalism

Why is it unprofessional to post about patients or workplace events on social media, even without names?

Answer

Details such as time, place or rare conditions can identify patients, breaching confidentiality. Posts also affect the profession's public trust, and can lead to employer discipline and regulator action.

#499 · Professionalism

You witness a coworker skipping a required safety or QC step repeatedly. What is your duty?

Answer

Report the unsafe practice to your supervisor or through the lab's reporting process, because patient and staff safety come first. Staying silent makes you part of the risk. Reports should be factual and documented.

#500 · Professionalism

A colleague asks about a friend's blood results. What is the correct response?

Answer

Decline to share them. Results are confidential and may be released only to the patient or authorized care providers under provincial privacy law, such as PHIPA in Ontario. Friendship or curiosity is not a valid reason.